

Artistverse
Software Design Development

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Bachelor of Science
Software Engineering Program

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Chapter 1 Introduction

1.1 Purpose and Scope

The purpose of the software design development is to present a complete software design of the Artistverse project. This document contains the outline of the software architecture, functions used, and activity diagrams with the user interface for this project.

1.2 User Characteristics

User	Role
Songwriter	A person who logs in to the Artistverse system and uses the system

1.3 Acronyms and Definitions

Acronyms

SRS	Software Requirement Specification
UI	User Interface
SDD	Software Design Document
FD	Function Description

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Definition

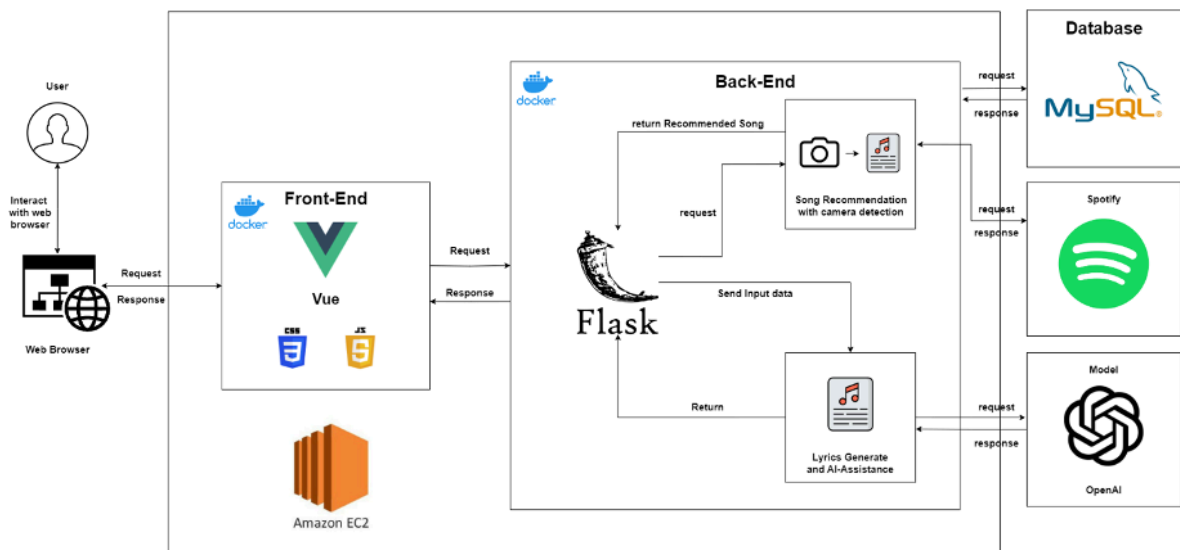
Name	Definition
IEEE	Institute for Electrical and Electronics Engineers. Biggest global interest group for engineers of different branches and computer scientists. [1]
Plan	A document series of tasks requires meeting an objective, typically including the associated schedule, budget, resources, organizational description and work breakdown structure. [1]
Feature	Feature Transformation of input parameters to output parameters based on a specified algorithm. It describes the functionality of the product in the language of the product. Used for requirements analysis, design, coding, testing or maintenance. [1]
Project Management	The application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed stakeholder needs and expectations from a project. [1]
Project Plan	A formal, approved document used to guide both project execution and project control. The primary uses of the project plan are to document planning assumptions and the decision, to facilitate communication among stakeholders, and to document approved scope, cost, and schedule baseline. [1]
Risk	An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives. It is a function of the probability of occurrence of a given threat's occurrence. [1]
Risk Management	The systematic application of management policies, procedures and practices to the tasks of identifying, analyzing, evaluating, treating and monitoring risk. [1]
System Testing	Testing conducted on a complete and integrated system for evaluate the system's compliance with its specified requirements. [1]
Unit Test	A test of individual programs or modules in order to remove a design or programming errors. [1]

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Traceability	The ability to trace the history, application or location of an item or activity, or work products or activities, by means of recorded identification. The establishment and maintenance of relationships between such items. Horizontal traceability describes the relationship between work products of the same type (e.g. Customer requirements). Vertical traceability describes the relationship between work products, which build or derived from each other (e.g., from customer requirements to qualification test cases). Bidirectional traceability allows to directly following relationships in both directions. [1]
Use Case	A use case is a description of a specific interaction between a system and its users to accomplish a particular goal or task. It outlines the sequence of steps or actions that a user performs in the system and describes the system's responses to those actions. Use cases help to identify and define the functional requirements of a system from the perspective of its users.

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Chapter 2 System Architecture



The user (Musician/Songwriter/Beginning songwriter) will be able to access the website front-end that was built by using Vue.js, a lightweight Python web framework called Flask, and Docker for containerization. The front-end allows users to create interactive web applications using reactive data binding and components. The back end enables us to create web applications by defining routes and handling requests and responses. Docker allows us to package our application and its dependencies into a single, portable container, ensuring easy deployment and running on any platform that supports Docker. We used the OpenAI model for the function-generated lyrics and Ai-Assistance to get the return result of the lyrics generated and the question-answer based on the user input. Spotify Api is for getting a list of the songs to use in the function song recommendation. The MySQL database stores input, lyrics, and other relevant information, which is used to retrieve song lyrics and other data when generating recommendations

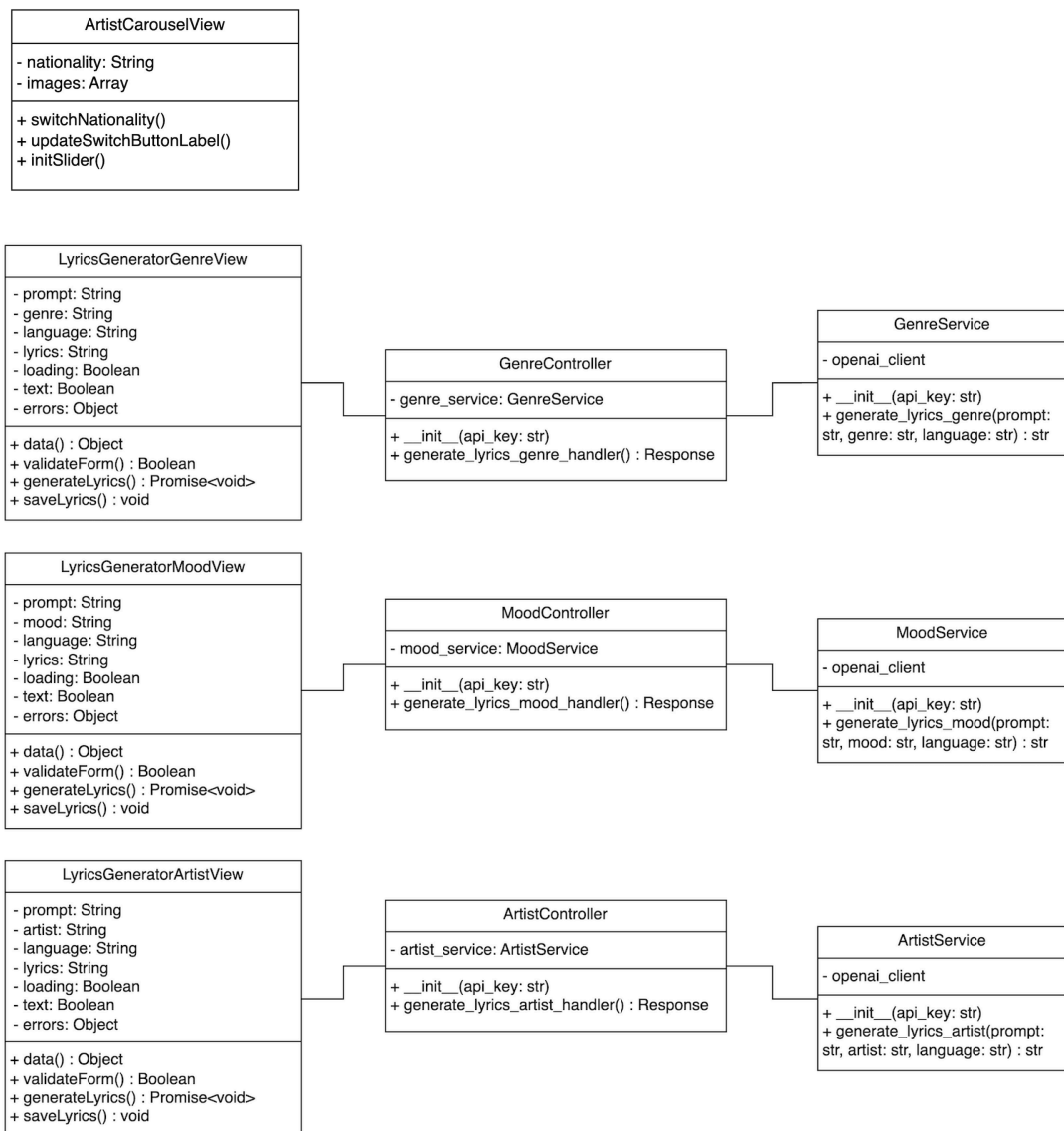
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Chapter 3 Detail Design

3.1 Progress I

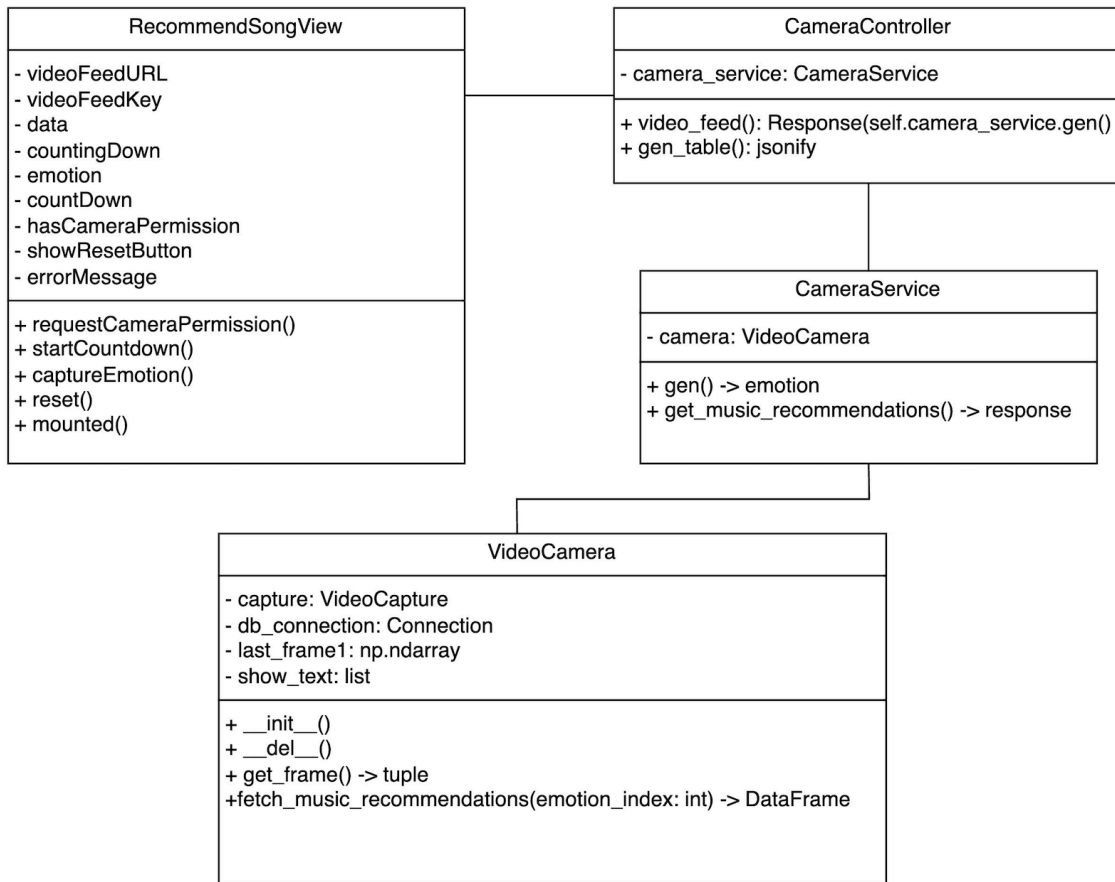
3.1.1 Class Diagram

3.1.1.1 Feature #1: Dynamic Lyric Composition



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3.1.1.2 Feature #2: Mood Detection Camera Integration



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3.1.2 Class Diagram Description

3.1.2.1 Views

3.1.2.1.1 Views: ArtistCarouselView

ArtistCarouselView
- nationality: String - images: Array
+ switchNationality() + updateSwitchButtonLabel() + initSlider()

Attributes:

ID	Name	Description	Type
1	Nationality	A string representing the current nationality of the artist.	String
2	Images	An array containing objects with information about artists' images including source, description, nationality, and popup status.	Array

Methods:

ID	Name	Description	Parameter	Return Type
1	switchNationality()	The function used to switch the display between Thai and international artists	selectedNationality (String): The new nationality to switch to.	void

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2	updateSwitchButtonLabel()	The function used to adjust the text on the display toggle button.	newLabel (String): The new label to set for the button.	void
3	initSlider()	Initializes the image slider for the artist's images	None	void

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3.1.2.1.2 Views: LyricsGeneratorGenreView

LyricsGeneratorGenreView
- prompt: String - genre: String - language: String - lyrics: String - loading: Boolean - text: Boolean - errors: Object
+ data() : Object + validateForm() : Boolean + generateLyrics() : Promise<void> + saveLyrics() : void

Attributes:

ID	Name	Description	Type
1	prompt	The topic or theme for the lyrics that the user inputs	String
2	genre	Dropdown selection for the genre of the lyrics	String
3	language	Dropdown selection for the language in which the lyrics are written	String
4	lyrics	Text area where the generated lyrics are displayed	String
5	loading	Boolean flag indicating if the lyrics generation is in progress	Boolean
6	text	Boolean flag indicating if text input is active	Boolean
7	errors	Object containing error messages related to the form inputs	Object

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Methods:

ID	Name	Description	Parameter	Return Type
1	data()	Retrieves the current state data of the form, including user inputs and errors	None	Object
2	validateForm()	Validates the form inputs to ensure all required fields are filled out correctly	None	Boolean
3	generateLyrics()	Initiates the lyrics generation process based on the provided form inputs	None	Promise<void>
4	saveLyrics()	Saves the generated lyrics to the songwriter's local computer in a .txt file format	None	void

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3.1.2.1.3 Views: LyricsGeneratorMoodView

LyricsGeneratorMoodView
- prompt: String - mood: String - language: String - lyrics: String - loading: Boolean - text: Boolean - errors: Object
+ data() : Object + validateForm() : Boolean + generateLyrics() : Promise<void> + saveLyrics() : void

Attributes:

ID	Name	Description	Type
1	prompt	The topic or theme for the lyrics that the user inputs	String
2	mood	Dropdown selection for the mood of the lyrics	String
3	language	Dropdown selection for the language in which the lyrics are written	String
4	lyrics	Text area where the generated lyrics are displayed	String
5	loading	Boolean flag indicating if the lyrics generation is in progress	Boolean
6	text	Boolean flag indicating if text input is active	Boolean
7	errors	Object containing error messages related to the form inputs	Object

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Methods:

ID	Name	Description	Parameter	Return Type
1	data()	Retrieves the current state data of the form, including user inputs and errors	None	Object
2	validateForm()	Validates the form inputs to ensure all required fields are filled out correctly	None	Boolean
3	generateLyrics()	Initiates the lyrics generation process based on the provided form inputs	None	Promise<void>
4	saveLyrics()	Saves the generated lyrics to the songwriter's local computer in a .txt file format	None	void

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3.1.2.1.4 Views: LyricsGeneratorArtistView

LyricsGeneratorArtistView
- prompt: String - artist: String - language: String - lyrics: String - loading: Boolean - text: Boolean - errors: Object
+ data() : Object + validateForm() : Boolean + generateLyrics() : Promise<void> + saveLyrics() : void

Attributes:

ID	Name	Description	Type
1	prompt	The topic or theme for the lyrics that the user inputs	String
2	artist	Dropdown selection for the artist style of the lyrics	String
3	language	Dropdown selection for the language in which the lyrics are written	String
4	lyrics	Text area where the generated lyrics are displayed	String
5	loading	Boolean flag indicating if the lyrics generation is in progress	Boolean
6	text	Boolean flag indicating if text input is active	Boolean
7	errors	Object containing error messages related to the form inputs	Object

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Methods:

ID	Name	Description	Parameter	Return Type
1	data()	Retrieves the current state data of the form, including user inputs and errors	None	Object
2	validateForm()	Validates the form inputs to ensure all required fields are filled out correctly	None	Boolean
3	generateLyrics()	Initiates the lyrics generation process based on the provided form inputs	None	Promise<void>
4	saveLyrics()	Saves the generated lyrics to the songwriter's local computer in a .txt file format	None	void

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3.1.2.1.5 Views: RecommendSongView

RecommendSongView
- videoFeedURL - videoFeedKey - data - countingDown - emotion - countDown - hasCameraPermission - showResetButton - errorMessage
+ requestCameraPermission() + startCountdown() + captureEmotion() + reset() + mounted()

Attributes:

ID	Name	Description	Type
1	videoFeedURL	The topic or theme for the lyrics that the user inputs	string
2	videoFeedKey	Key for the video feed	int
3	data	List of song recommendations	array
4	countingDown	Indicates if countdown is active	boolean
5	emotion	Detected emotion	string
6	countDown	Countdown timer value	int
7	hasCameraPermission	Indicates if camera permission is granted	boolean
8	showResetButton	Indicates if reset button should be shown	boolean

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9	errorMessage	Error message for mood detection	string
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Methods:

ID	Name	Description	Parameter	Return Type
1	requestCameraPermission	For requests camera permission	None	void
2	startCountdown	Starts the countdown for emotion detection	None	void
3	captureEmotion	Captures the Songwriter emotion and fetches song recommendations	None	void
4	reset	Resets the state	None	void
5	mounted	Lifecycle hook to request camera permission on mount	None	void

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3.1.2.2 Controller

3.1.2.2.1 Controller: GenreController

GenreController
- genre_service: GenreService
+ __init__(api_key: str) + generate_lyrics_genre_handler() : Response

Attributes:

ID	Name	Description	Type
1	genre_service	An instance of GenreService used to handle operations related to genres	GenreService

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initializes the controller with the necessary API key to access the genre service	api_key: str	Response
2	generate_lyrics_genre_handler()	Handles the request to generate lyrics based on the provided genre	None	Response

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3.1.2.2.2 Controller: MoodController

MoodController
- mood_service: MoodService
+ __init__(api_key: str) + generate_lyrics_mood_handler() : Response

Attributes:

ID	Name	Description	Type
1	mood_service	An instance of MoodService used to handle operations related to moods	MoodService

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initializes the controller with the necessary API key to access the mood service	api_key: str	Response
2	generate_lyrics_mood_handler()	Handles the request to generate lyrics based on the provided mood	None	Response

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3.1.2.2.2 Controller: ArtistController

ArtistController
- artist_service: ArtistService
+ __init__(api_key: str) + generate_lyrics_artist_handler() : Response

Attributes:

ID	Name	Description	Type
1	artist_service	An instance of ArtistService used to handle operations related to artists	ArtistService

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initializes the controller with the necessary API key to access the artist service	api_key: str	Response
2	generate_lyrics_artist_handler()	Handles the request to generate lyrics based on the provided artist style	None	Response

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3.1.2.2.4 Controller: CameraController

CameraController
- camera_service: CameraService
+ video_feed(): Response(self.camera_service.gen()) + gen_table(): jsonify

Attributes:

ID	Name	Description	Type
1	camera_service	Instance of CameraService	CameraService

Methods:

ID	Name	Description	Parameter	Return Type
1	video_feed	Provides the video feed response	None	Response(self.camera_service.gen())
2	gen_table	Generates a JSON response with music recommendations	None	jsonify

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3.1.2.3 Service

3.1.2.3.1 Service: GenreService

GenreService
- openai_client
+ __init__(api_key: str) + generate_lyrics_genre(prompt: str, genre: str, language: str) : str

Attributes:

ID	Name	Description	Type
1	openai_client	Client for OpenAI API	OpenAI Client

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initialize with API key	api_key: str	None
2	generate_lyrics_genre(prompt: str, genre: str, language: str)	Generate lyrics based on genre	prompt: str, genre: str, language: str	str

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3.1.2.3.2 Service: MoodService

MoodService
- openai_client
+ __init__(api_key: str) + generate_lyrics_mood(prompt: str, mood: str, language: str) : str

Attributes:

ID	Name	Description	Type
1	openai_client	Client for OpenAI API	OpenAI Client

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initialize with API key	api_key: str	None
2	generate_lyrics_mood(prompt: str, mood: str, language: str)	Generate lyrics based on mood	prompt: str, mood: str, language: str	str

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3.1.2.3.3 Service: ArtistService

ArtistService
- openai_client
+ __init__(api_key: str) + generate_lyrics_artist(prompt: str, artist: str, language: str) : str

Attributes:

ID	Name	Description	Type
1	openai_client	Client for OpenAI API	OpenAI Client

Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initialize with API key	api_key: str	None
2	generate_lyrics_artist(prompt: str, artist: str, language: str)	Generate lyrics based on artist style	prompt: str, artist: str, language: str	str

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3.1.2.2.4 Service: CameraService

CameraService
- camera: VideoCamera
+ gen() -> emotion + get_music_recommendations() -> response

Attributes:

ID	Name	Description	Type
1	camera	Instance of VideoCamera	VideoCamera

Methods:

ID	Name	Description	Parameter	Return Type
1	gen	Generates video frames with emotion detection	None	emotion
2	get_music_recommendations	Fetches music recommendations based on detected emotions	None	response

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3.1.2.4 Model

3.1.2.4.1 Model: VideoCamera

VideoCamera
- capture: VideoCapture - db_connection: Connection - last_frame1: np.ndarray - show_text: list
+ __init__() + __del__() + get_frame() -> tuple + fetch_music_recommendations(emotion_index: int) -> DataFrame

Attributes:

ID	Name	Description	Type
1	capture	VideoCapture object for capturing video frames	VideoCapture
2	db_connection	This attribute handles the database connection for fetching music recommendations.	Connection
3	last_frame1	Last captured video frame	np.ndarray
4	show_text	List to store detected emotion index	list

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Methods:

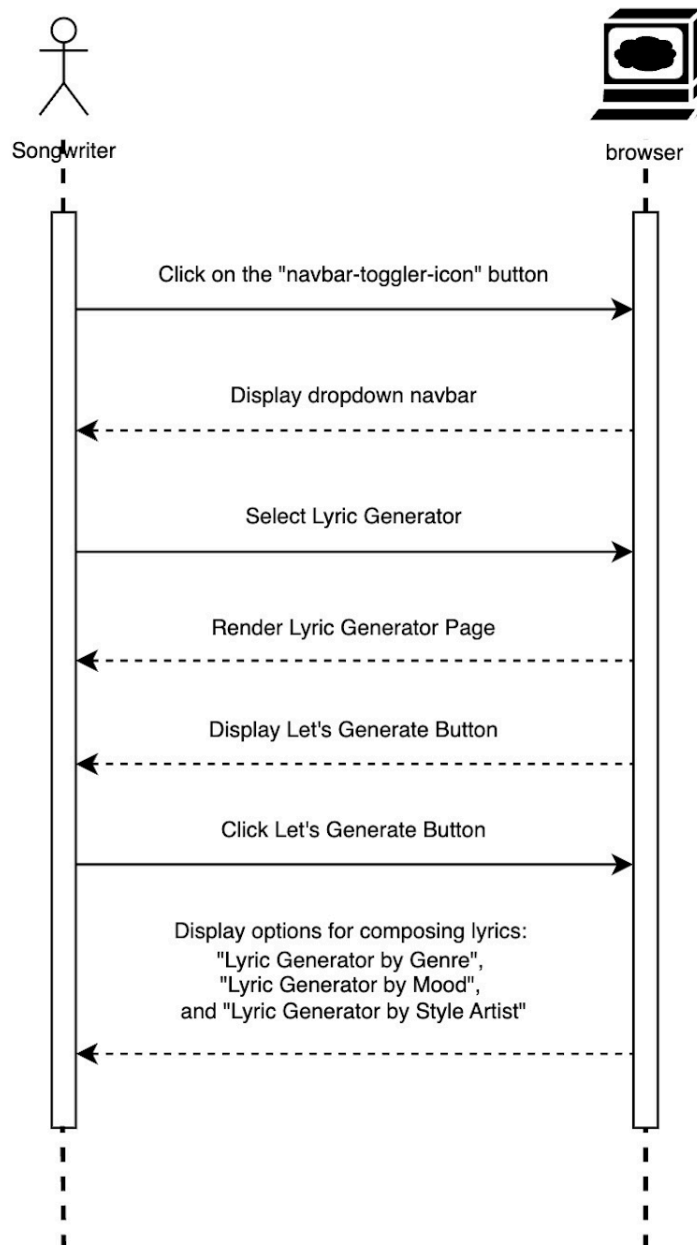
ID	Name	Description	Parameter	Return Type
1	init	This method initializes the video capture and database connection.	None	None
2	del	This method releases the video capture and closes the database connection.	None	None
3	get_frame	This method captures frames from the webcam and predicts the emotion.	None	Tuple (image bytes, DataFrame, emotion string)
4	fetch_music_recommendations	This method fetches music recommendations based on the detected emotion.	emotion_index (int)	DataFrame

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3.1.3 Sequence Diagram

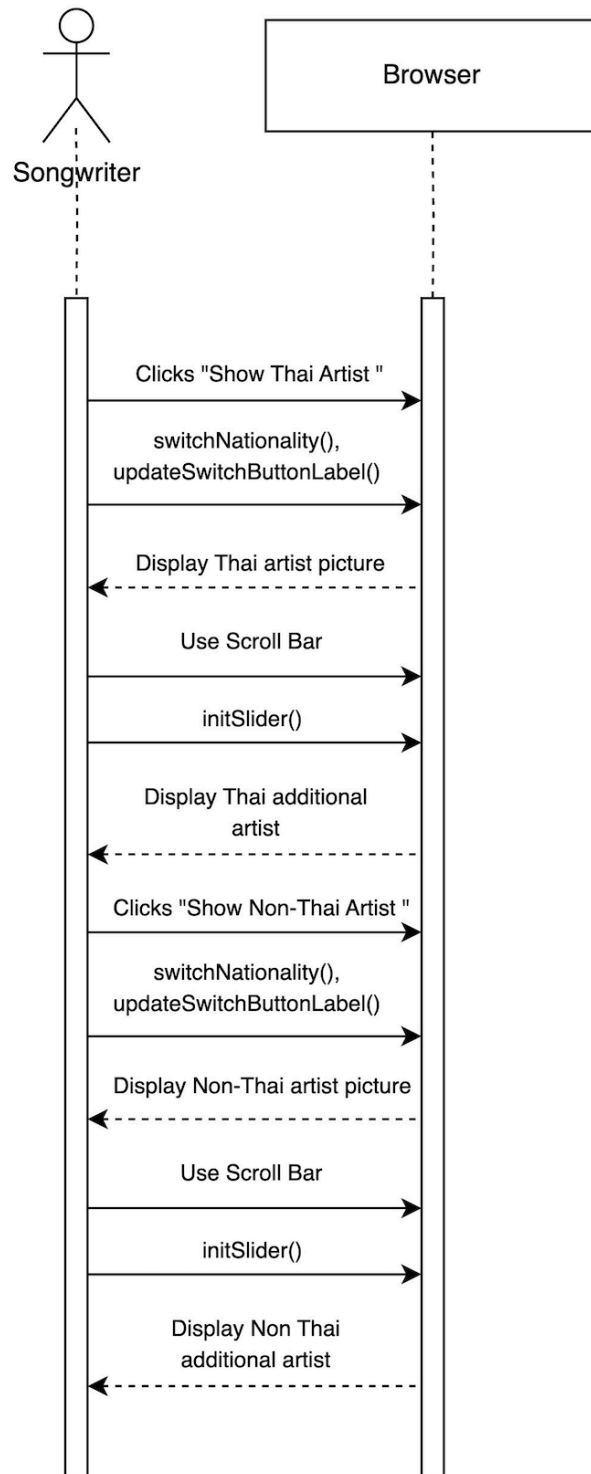
Progress I

3.1.3.1 SD 01 Access Lyric Generation Options



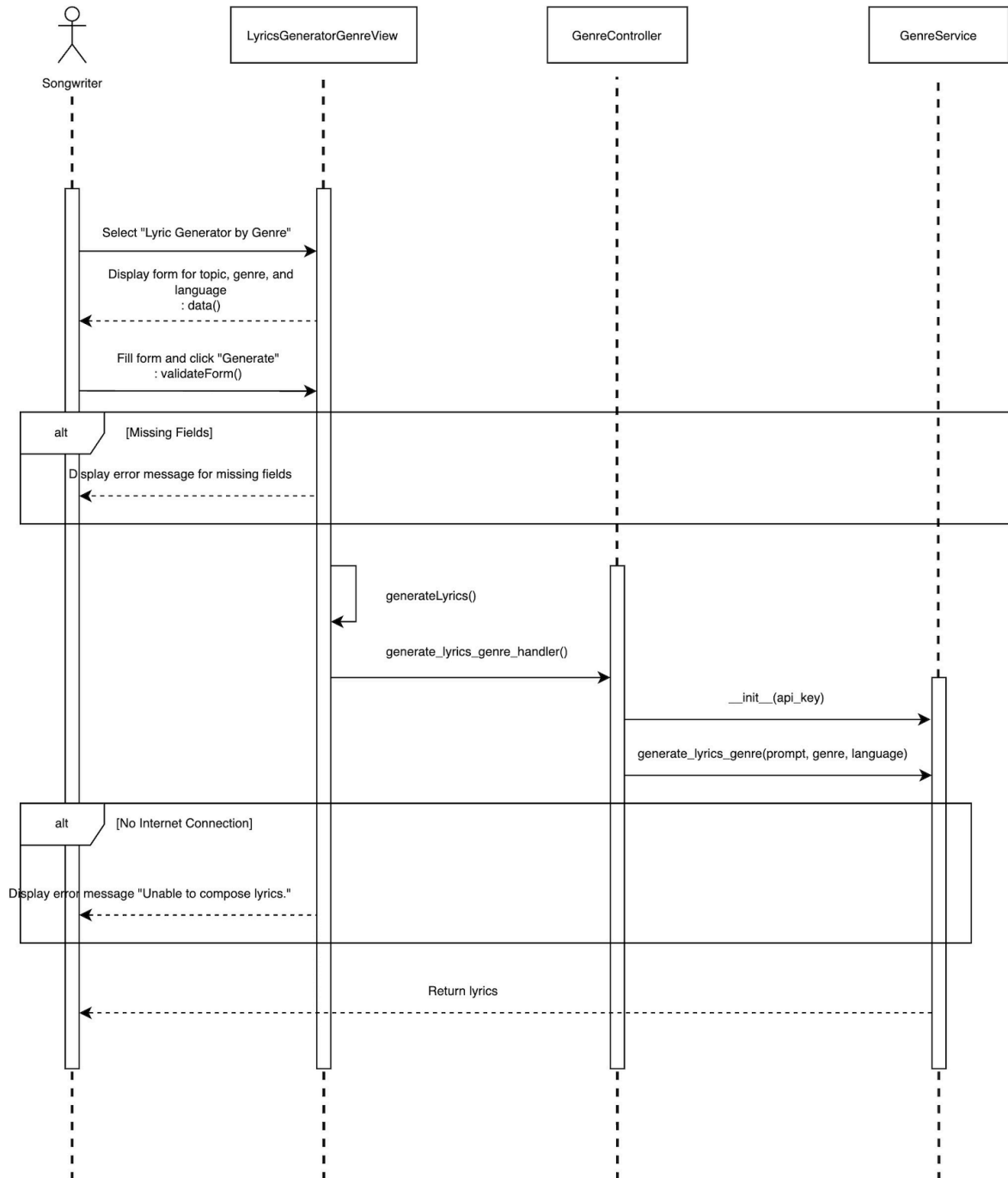
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3.1.3.2 SD 02 View Artist Information



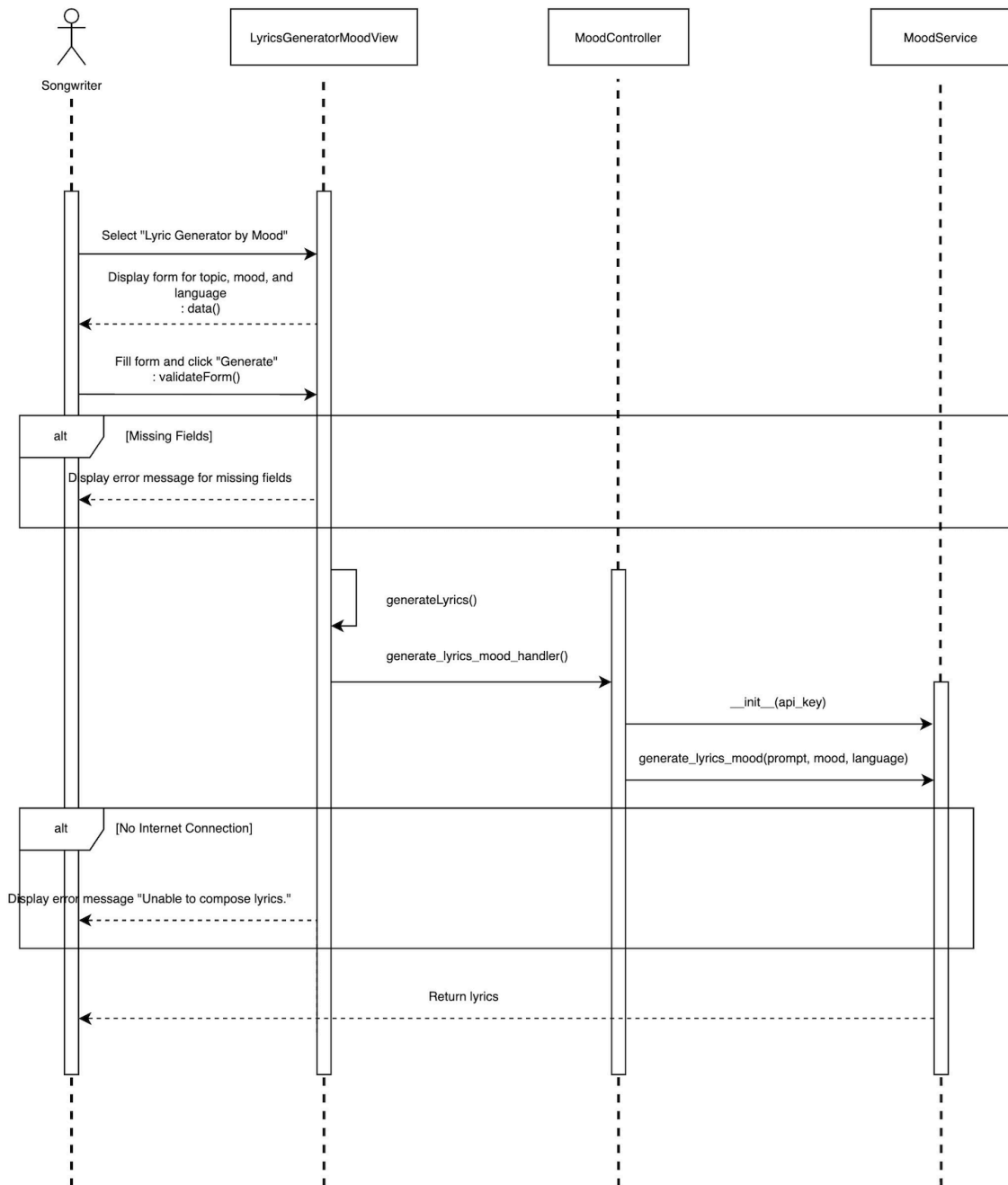
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3.1.3.3 SD 03 Compose Lyrics by Genre



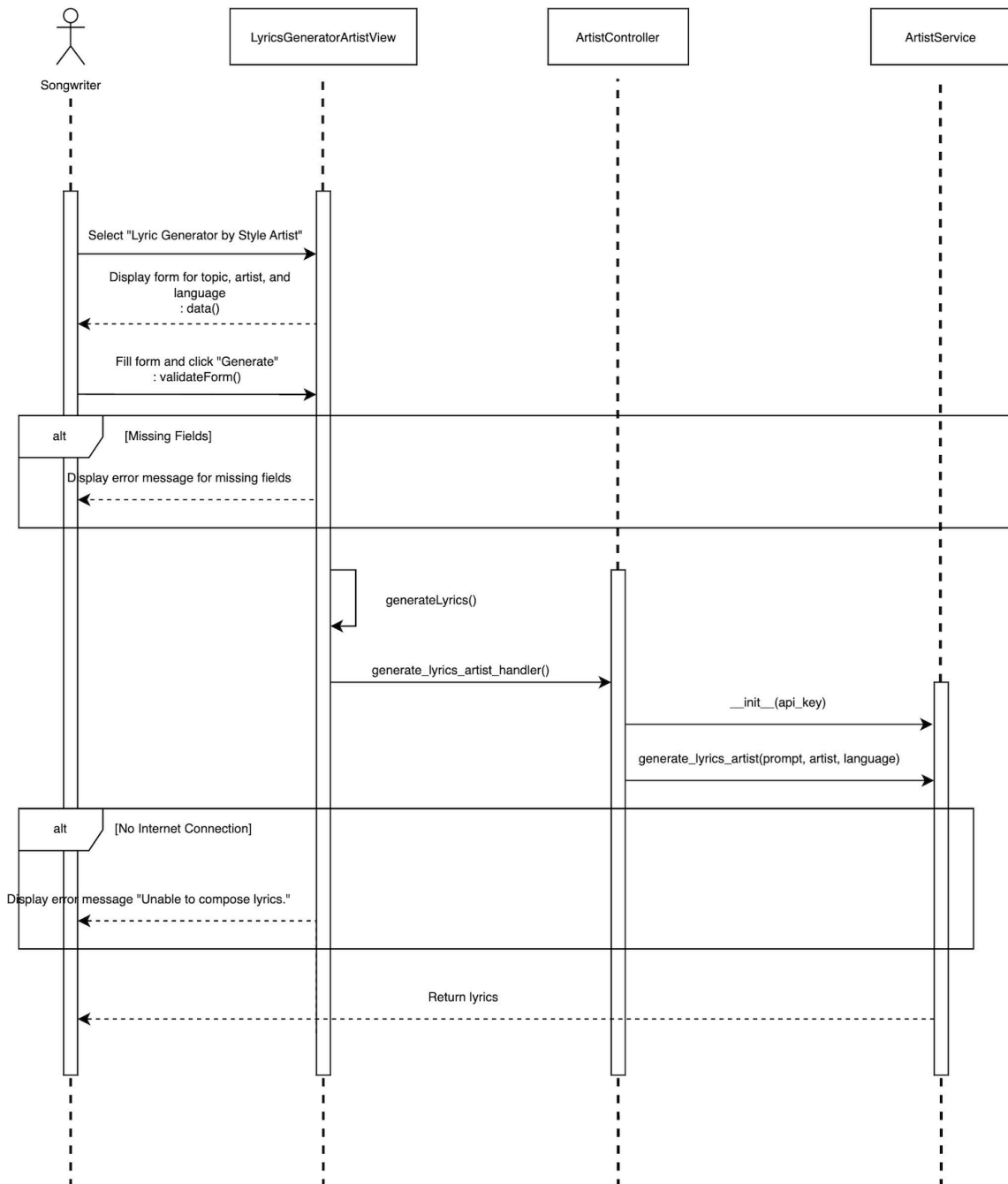
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3.1.3.4 SD 04 Compose Lyrics by Mood



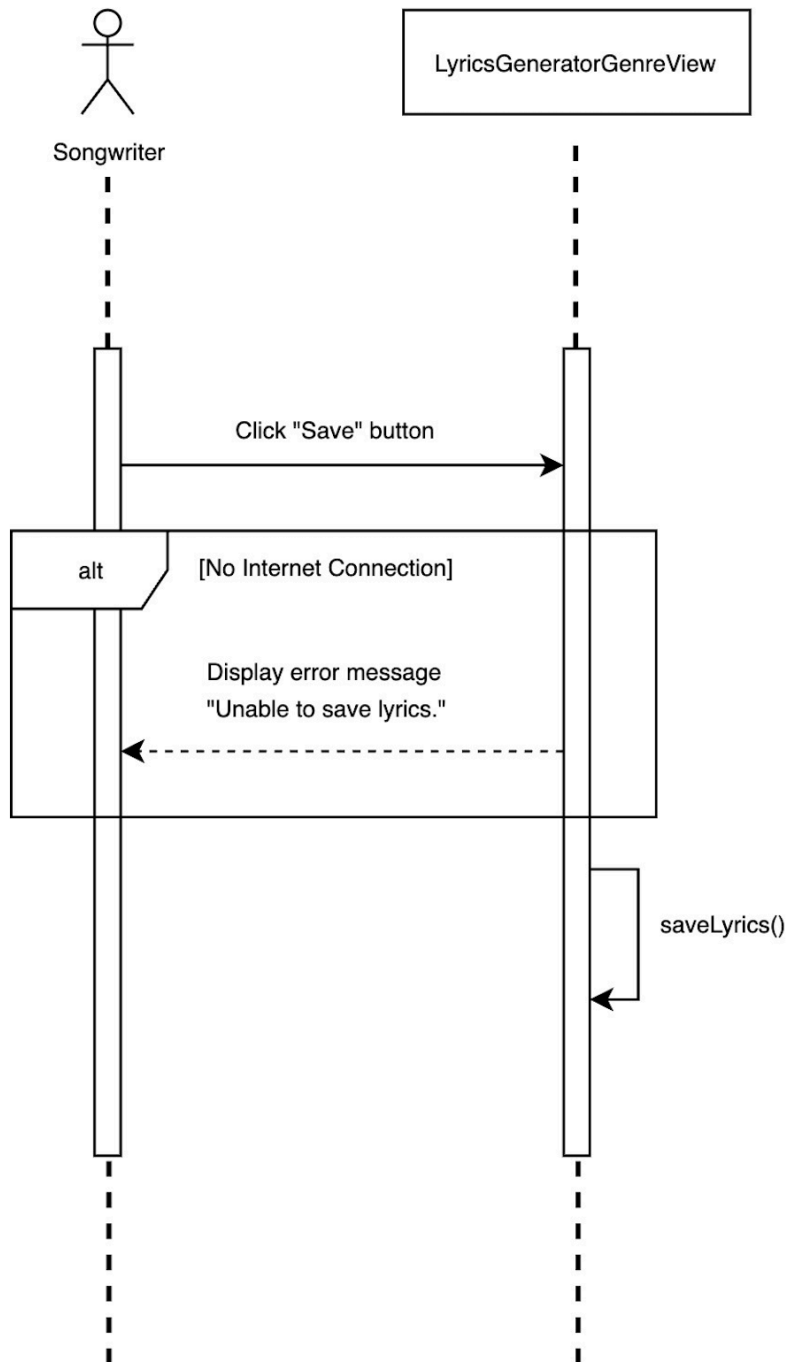
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3.1.3.5 SD 05 Compose Lyrics by Style Artist

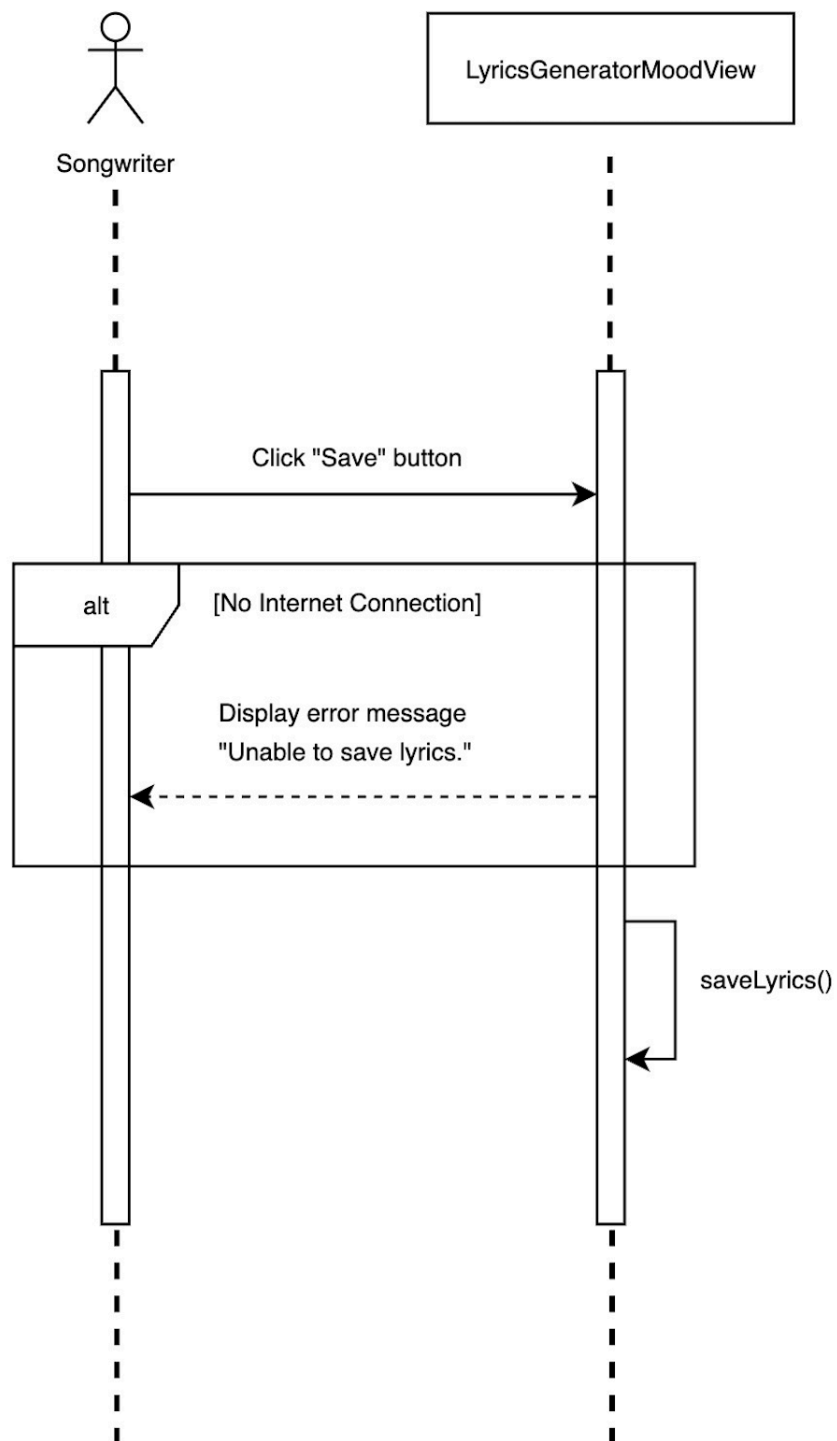


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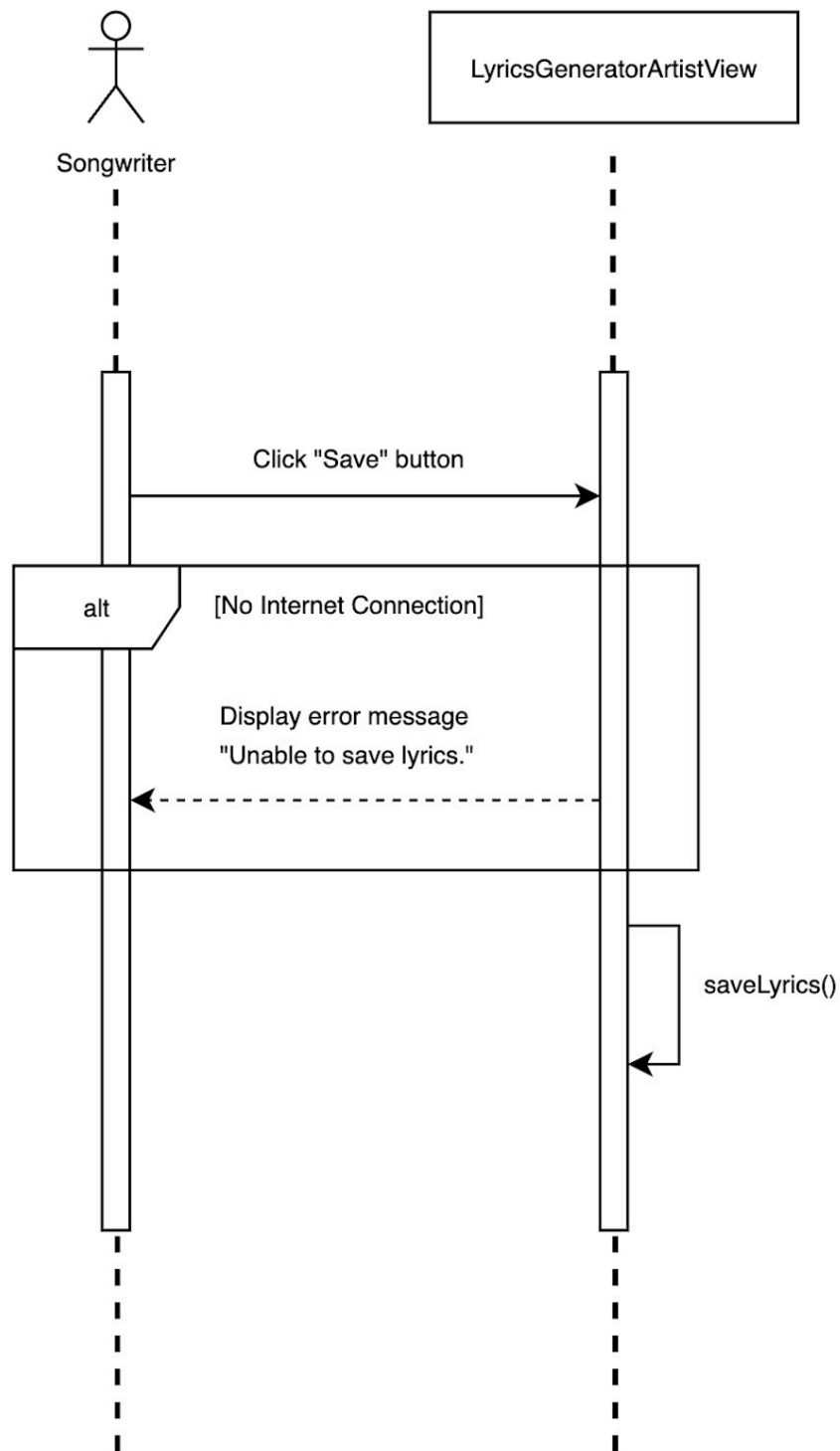
3.1.3.6 SD 06 Save Lyrics



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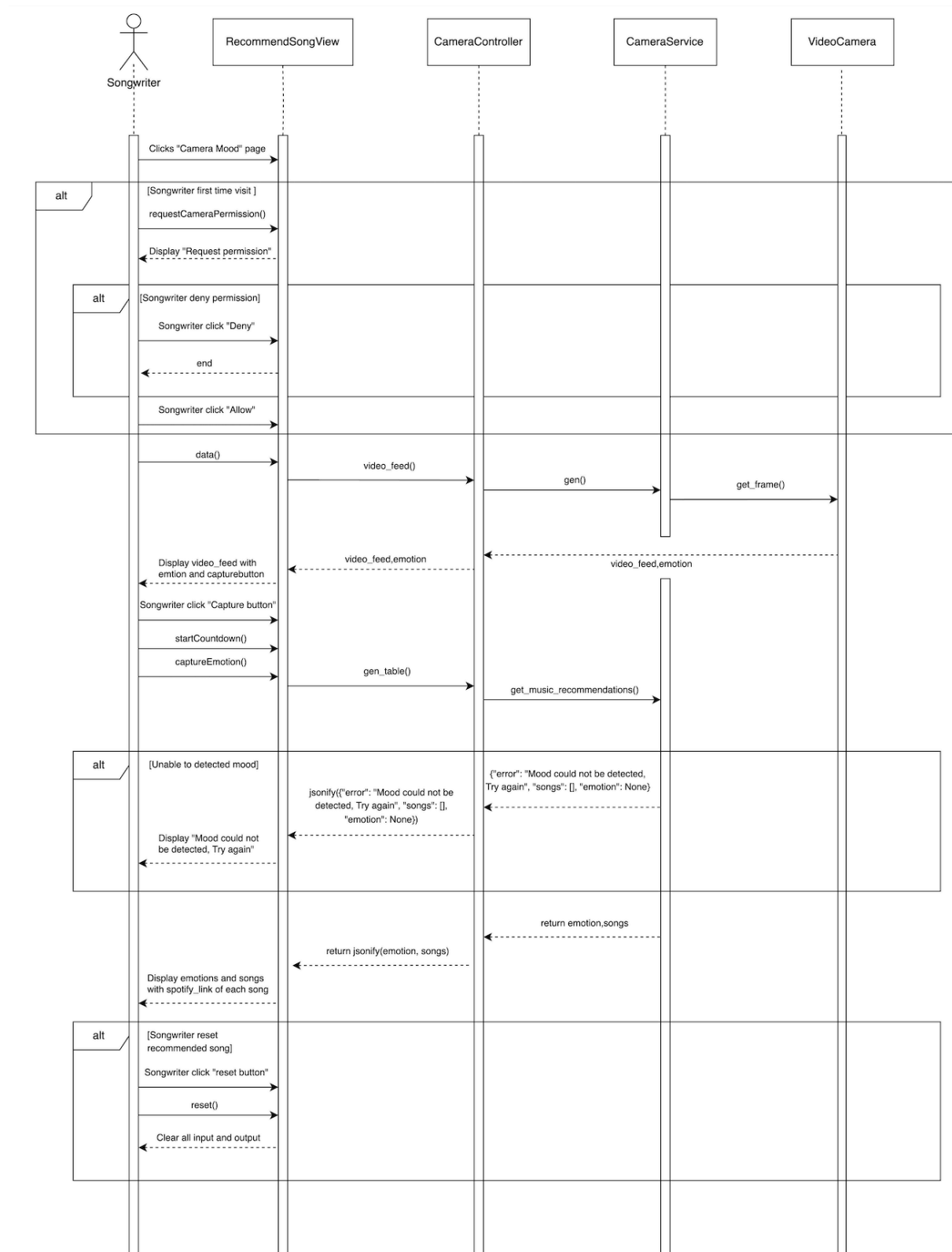


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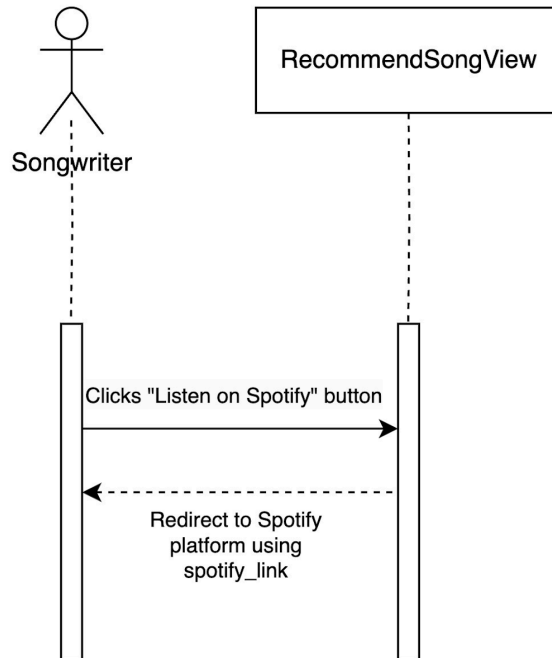
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3.1.3.7 SD 07 Capture facial expression and Recommended song



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3.1.3.8 SD 08 Select the song from the recommended playlist to redirected to Spotify

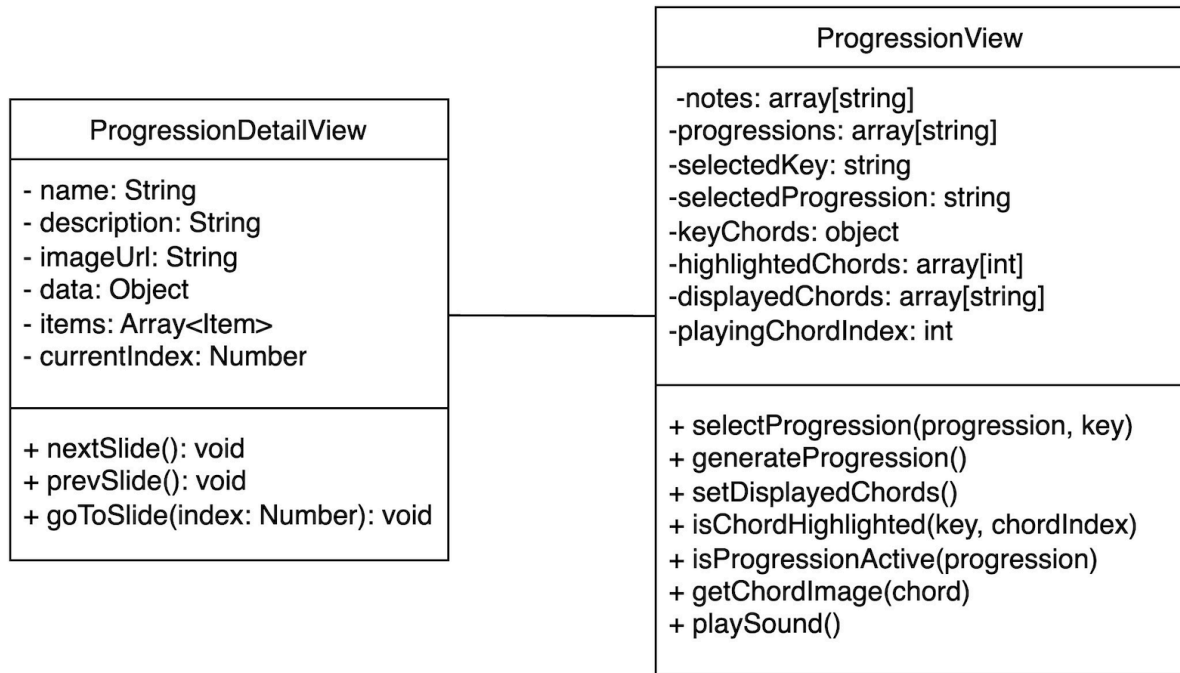


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3.2 Progress II

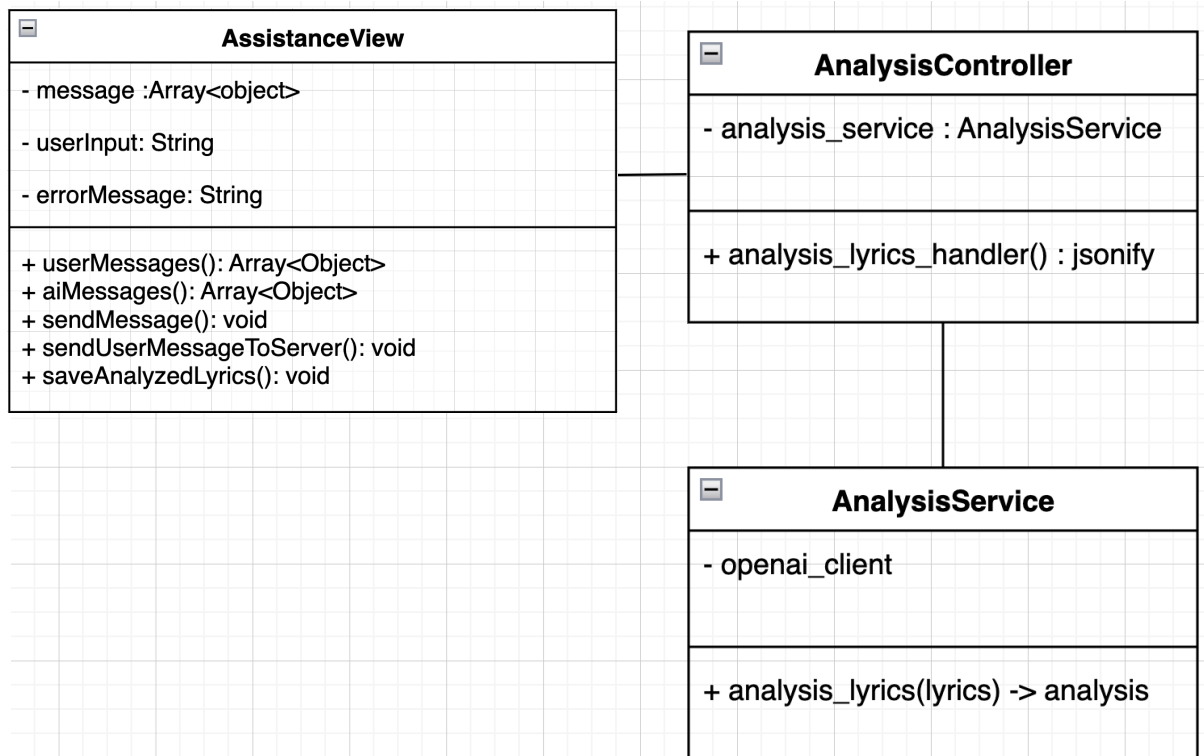
3.2.1 Class Diagram

3.2.1.1 Feature #3: Chord Progression



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3.2.1.2 Feature #4: Ai Lyrics Analysis

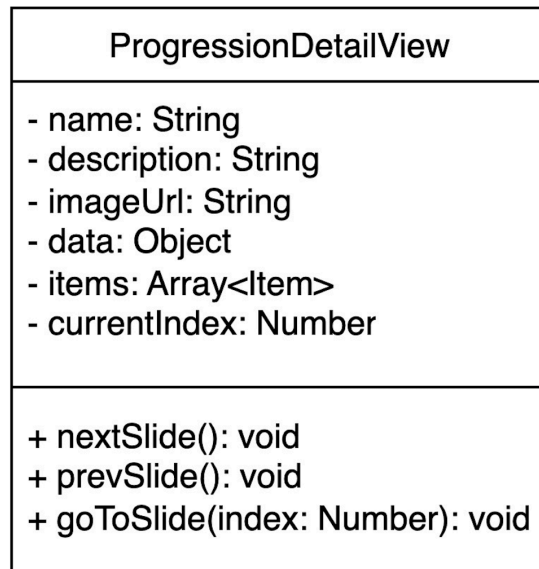


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3.2.2 Class Diagram Description

3.2.2.1 Views

3.2.2.1.1 Views: ProgressionDetailView



Attributes:

ID	Name	Description	Type
1	name	This attribute stores the name of the progression or view.	String
2	description	This attribute holds a brief description or explanation of the progression.	String
3	imageUrl	This attribute contains the URL of an image associated with the progression.	String
4	data	This attribute stores additional information or metadata related to the progression.	Object
5	items	This attribute is an array that	Array<Item>

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		contains items related to the progression, such as steps or details.	
6	currentIndex	This attribute tracks the current index of the displayed item or slide in the progression.	Name

Methods:

ID	Name	Description	Parameter	Return Type
1	nextSlide()	This method advances to the next slide or item in the progression view.	None	void
2	prevSlide()	This method moves back to the previous slide or item.	None	void
3	goToSlide(index: Number)	This method navigates directly to a specific slide or item based on the provided index.	index: Number	void

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3.2.2.1.2 Views: ProgressionView

ProgressionView
-notes: array[string] -progressions: array[string] -selectedKey: string -selectedProgression: string -keyChords: object -highlightedChords: array[int] -displayedChords: array[string] -playingChordIndex: int
+ selectProgression(progression, key) + generateProgression() + setDisplayedChords() + isChordHighlighted(key, chordIndex) + isProgressionActive(progression) + getChordImage(chord) + playSound()

Attributes:

ID	Name	Description	Type
1	notes	This attribute is an array that stores musical notes relevant to the progression.	String
2	progressions	This attribute is an array that stores different chord progressions.	Array[String]
3	selectedKey	This attribute stores the currently selected musical key.	String
4	selectedProgression	This attribute stores the currently selected chord progression.	String
5	keyChords	This attribute stores an object that maps each key to its corresponding chords.	object

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6	highlightedChords	This attribute stores an array of indices indicating which chords are highlighted in the current progression.	Array[int]
7	displayedChords	This attribute is an array that stores the chords currently displayed based on the selected key and progression.	Array[String]
8	playingChordIndex	This attribute stores the index of the chord that is currently being played.	int

Methods:

ID	Name	Description	Parameter	Return Type
1	selectProgression (progression, key)	This method selects a specific chord progression and musical key.	progression: String key: String	void
2	generateProgression()	This method generates the chord progression based on the selected key.	None	void
3	setDisplayChords()	This method updates the displayedChords array based on the selected progression and key.	None	void
4	isChordHighlighted(key, chordIndex)	This method checks if a specific chord is highlighted.	key: String chordIndex: Number	Boolean
5	isProgressionActive(progression)	This method checks if a specific progression is currently active.	progression: String	Boolean

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6	getChordImage(chord)	This method retrieves the image associated with a specific chord.	chord: String	String (The URL or path to the image)
7	playSound()	This method plays the sound of the currently selected or highlighted chord.	None	void

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3.2.2.1.1 Views: AssistanceView

AssistanceView
- message :Array<object> - userInput: String - errorMessage: String
+ userMessages(): Array<Object> + aiMessages(): Array<Object> + sendMessage(): void + sendUserMessageToServer(): void + saveAnalyzedLyrics(): void

Attributes:

ID	Name	Description	Type
1	messages	Stores a list of messages, where each message has an id (number), a role (string, either "user" or "ai"), and content (string) representing the text of the message.	Array<{ id: number, role: string, content: string }>
2	userInput	Holds the current input text from the user, which is bound to the input field in the template.	String
3	errorMessage	Holds any error messages to be displayed if there are issues with user input or network connectivity.	String

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Methods:

ID	Name	Description	Parameter	Return Type
1	sendMessage()	Checks if the userInput is not empty, pushes the message to the messages array, and calls sendUserMessageToServer(). Then it clears the userInput.	None	void
2	sendUserMessageToServer()	Sends the userInput to the server via a POST request and updates the messages array with the AI's response.	None	void
3	saveAnalyzedLyrics()	Checks if there is an internet connection and whether any analyzed lyrics are present. If checks are passed, it concatenates all AI messages and initiates a download of the analyzed lyrics as a text file. Handles errors or issues in the process.	None	void
4	userMessages()	Filters and returns messages from the messages array that have a role of "user".	None	Array<{ id: number, role: string, content: string }>
5	aiMessage()	Filters and returns messages from the messages array that have a role of "ai".	None	Array<{ id: number, role: string, content: string }>

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3.2.2.2 Controller

3.2.2.2.1 Controller: AnalysisController

 AnalysisController
- analysis_service : AnalysisService
+ analysis_lyrics_handler() : jsonify

Attributes:

ID	Name	Description	Type
1	analysis_service	An instance of the AnalysisService class used to perform the lyrics analysis.	AnalysisService

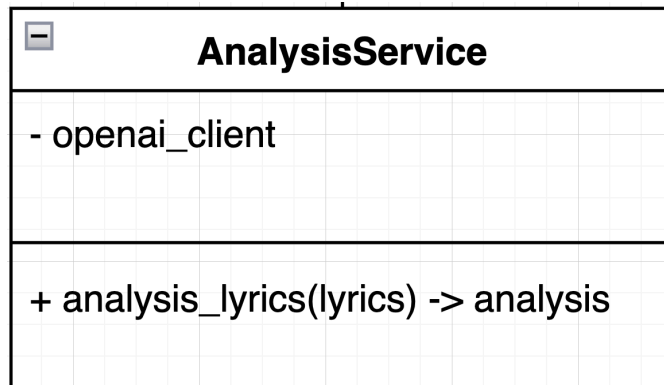
Methods:

ID	Name	Description	Parameter	Return Type
1	init(api_key: str)	Initializes the controller with the necessary API key to access the genre service	api_key: str	Response
2	analysis_lyrics_handler()	Receives lyrics from a JSON request, passes them to the AnalysisService for processing, and returns a JSON response with the analysis result.	None	Response

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3.2.2.3 Service

3.2.2.3.1 Service: AnalysisService



Attributes:

ID	Name	Description	Type
1	openai_client	An instance of the OpenAI API client used to communicate with the OpenAI API.	OpenAI Client

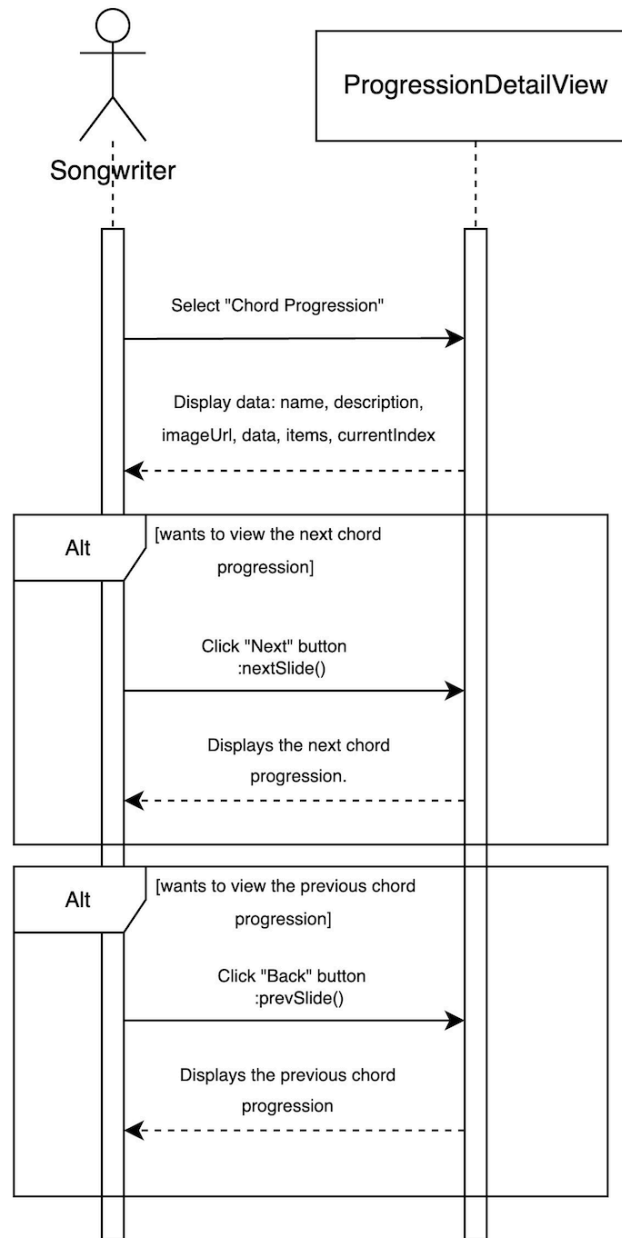
Methods:

ID	Name	Description	Parameter	Return Type
1	analysis_lyrics(lyrics)	Takes lyrics as a string, sends them to the OpenAI API, and returns a string containing the AI-generated analysis of the lyrics.	lyrics: str	String

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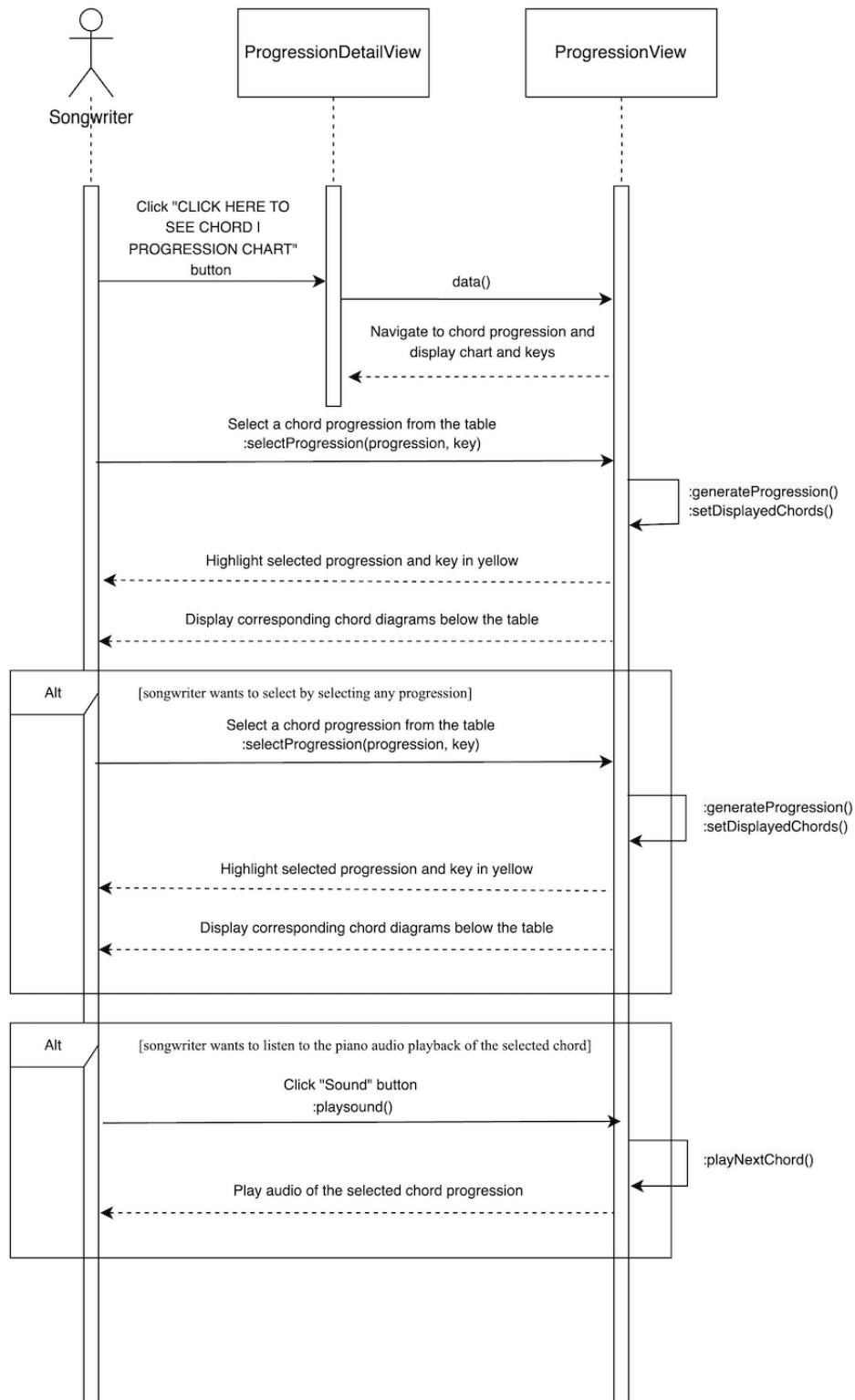
3.2.3 Sequence Diagram

3.2.3.9 SD 09 Access and View Chord Progression Information



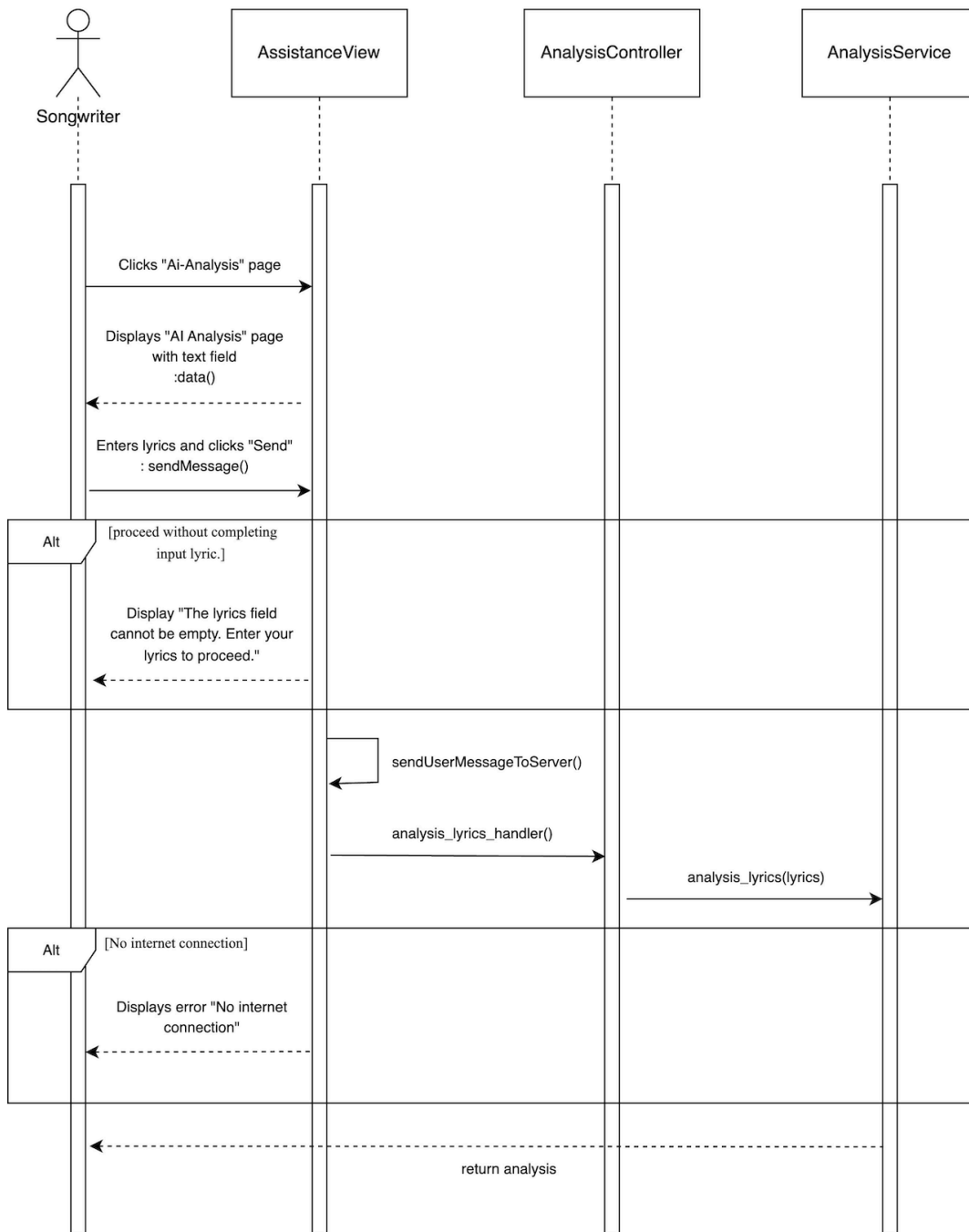
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3.2.3.10 SD 10 Chord Progression Chart Interaction



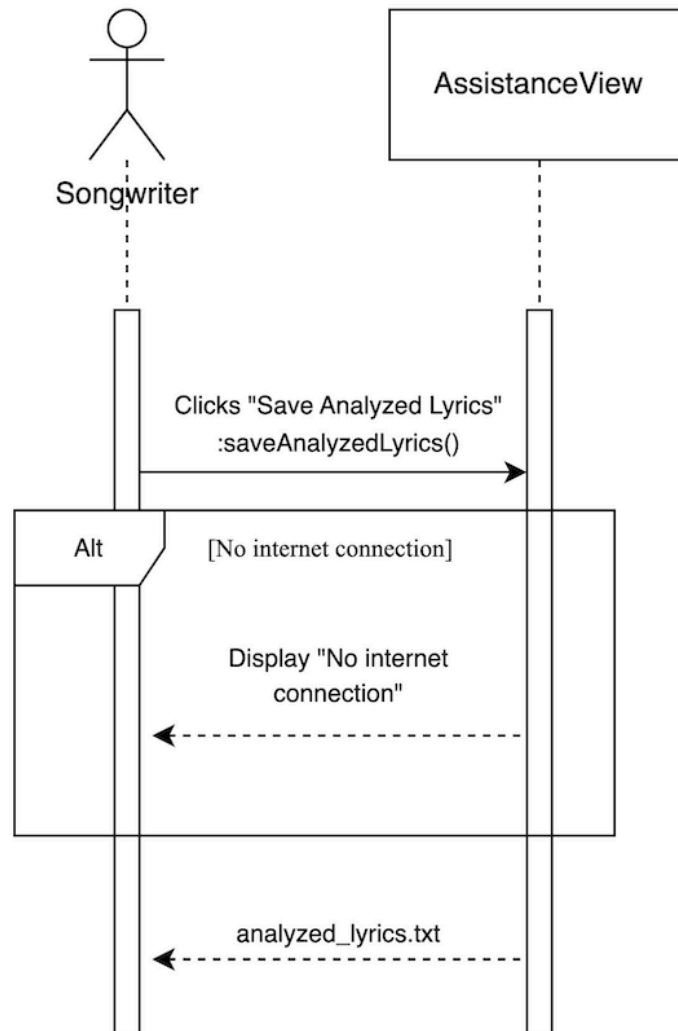
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3.2.3.11 SD 11 Input Lyrics and View Musical Analysis



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3.2.3.12 SD 12 Save Analyzed Lyrics



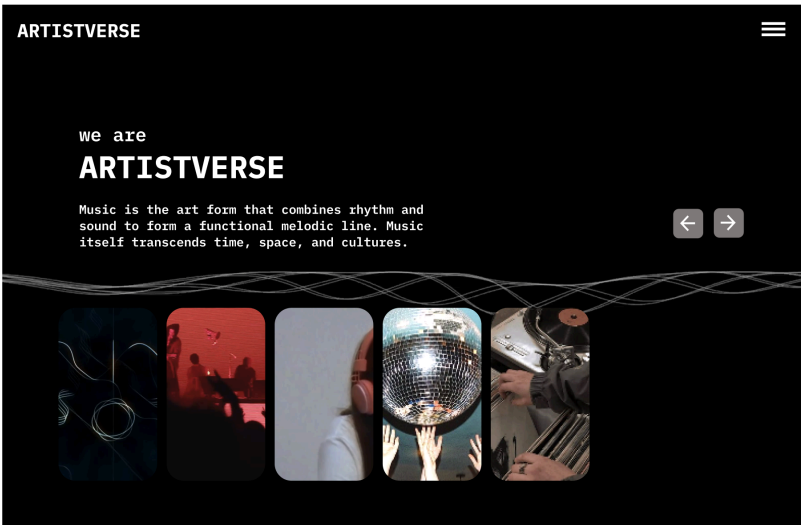
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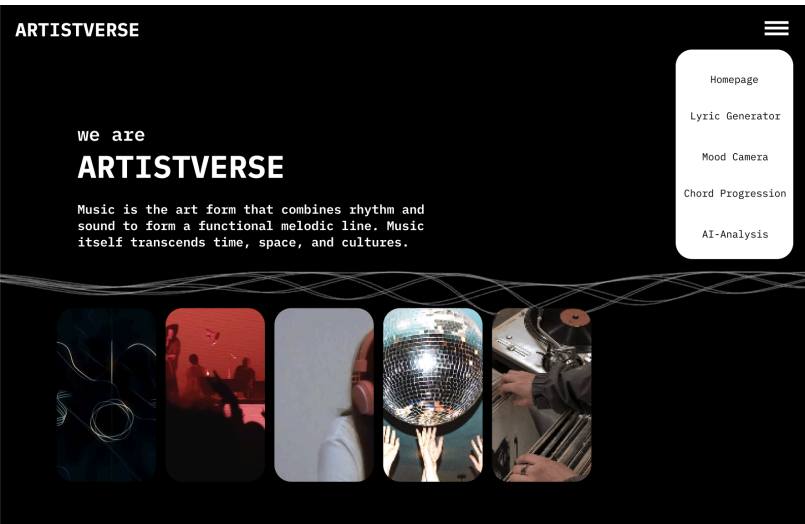
4.1 Progress I

4.1.1 Web Application UI

UI-01 Home page



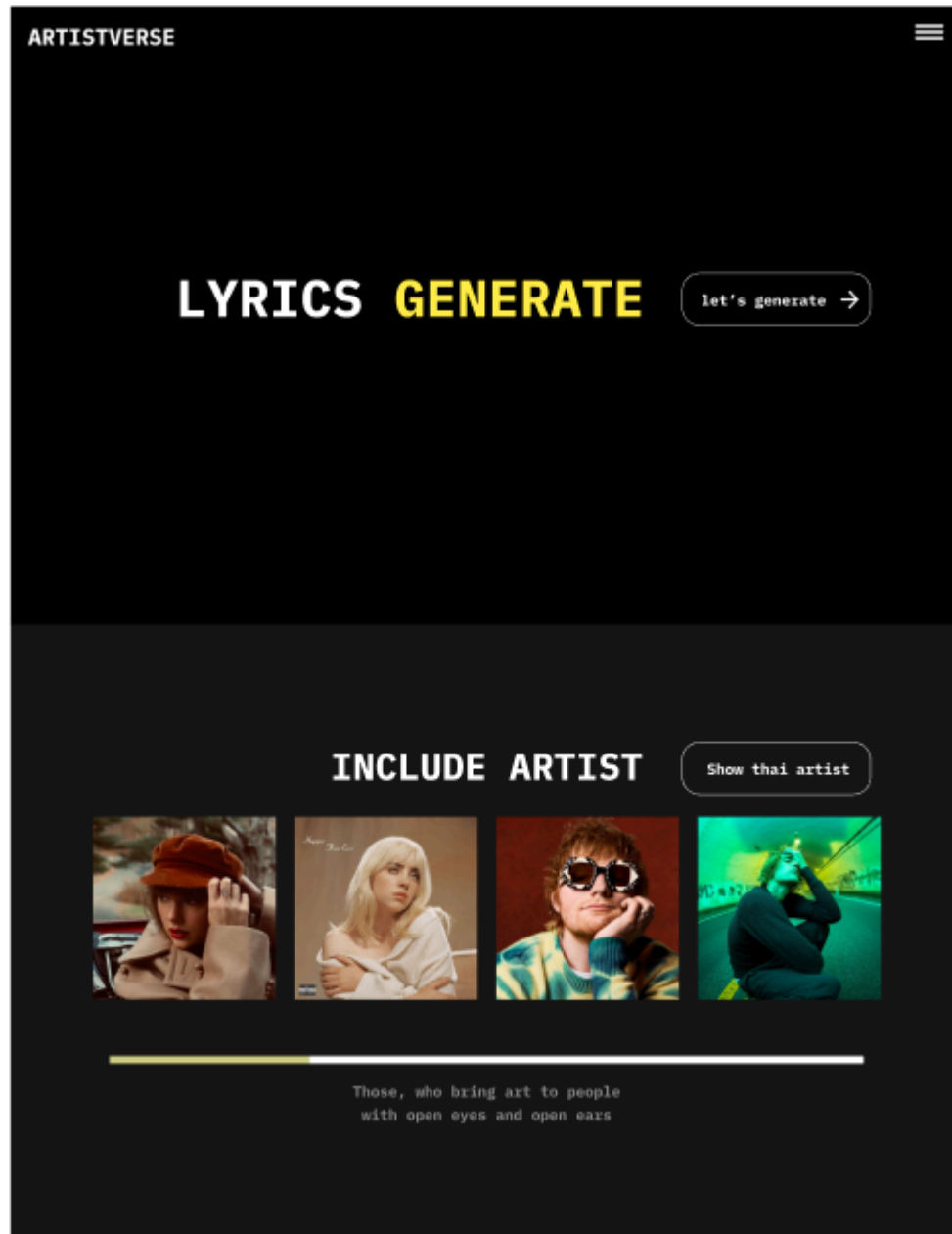
UI-02 Home page with "navbar-toggler-icon" button



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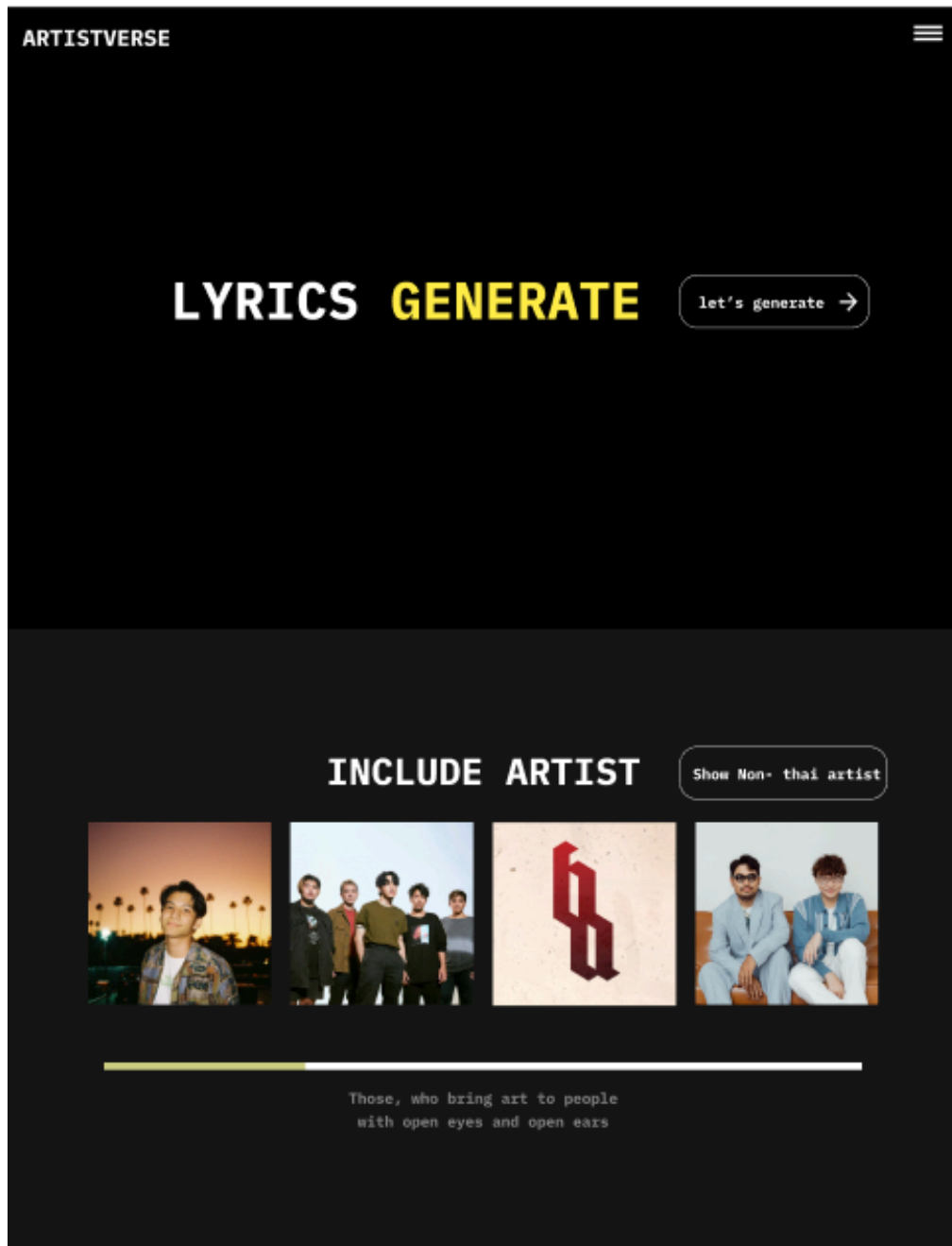
UI-03 Lyric Generator page with "Let's generate" button that can redirect the Songwriter to the lyrics generator by Genre page and information about including the Non-thai artist.

X



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UI-03-1 Lyric Generator page with "Let's generate" button that can redirect the Songwriter to the lyrics generator by Genre page and information about including the Thai artist.



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UI-04 Lyric Generator by Genre Page

ARTISTVERSE

Lyrics Generator by "Genre"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

Your lyrics topic

Genre: Language:

GENERATE

RESULT

UI-04-1 With invalid input

ARTISTVERSE

Lyrics Generator by "Genre"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

* Please input a lyric topic

Genre: Language:

* Please select Genre

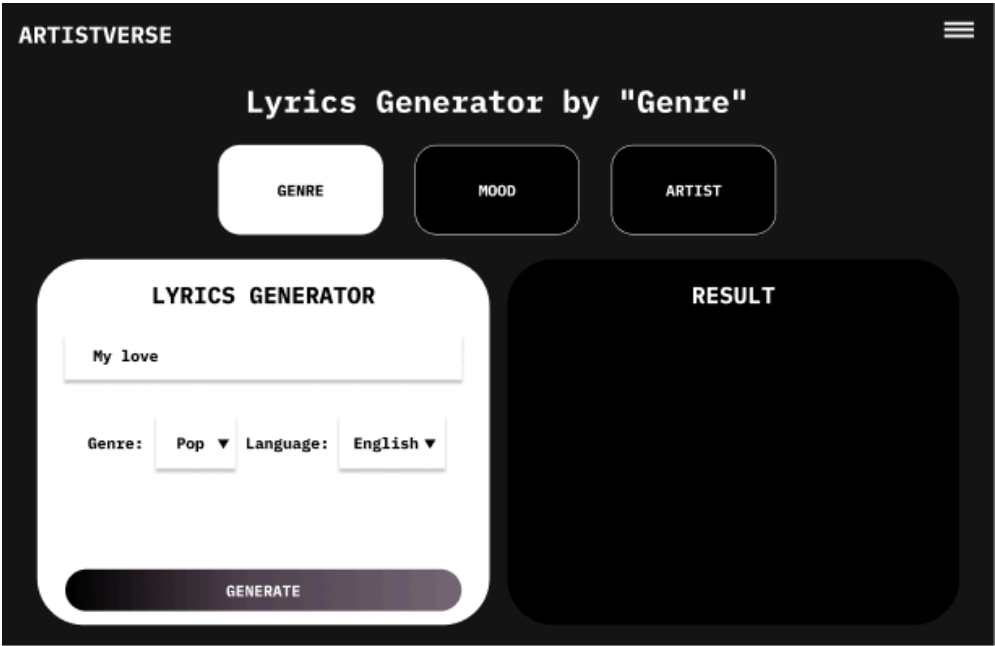
* Please select language

GENERATE

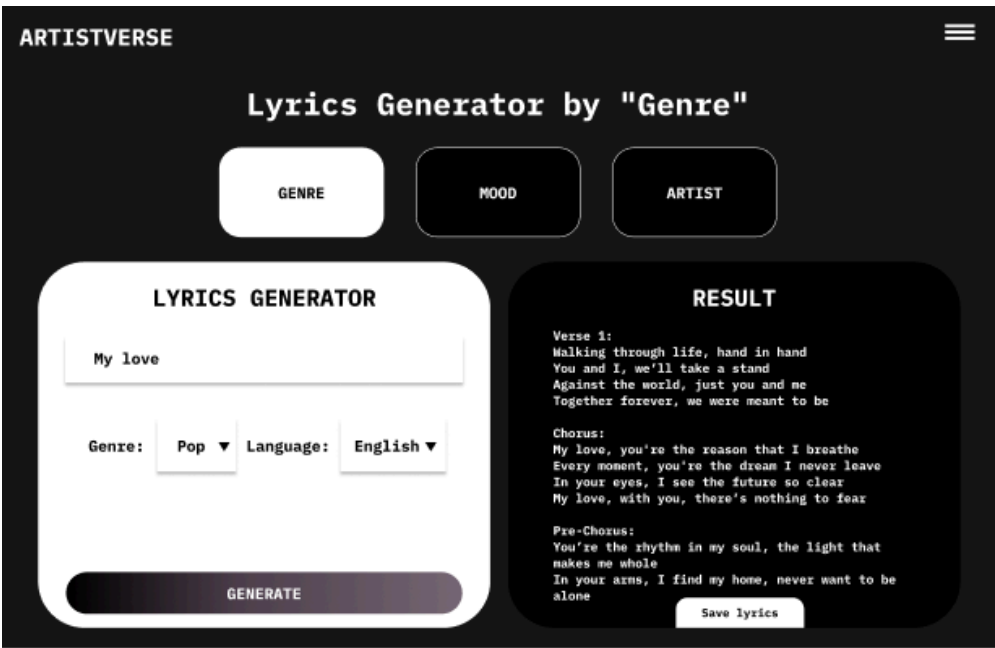
RESULT

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UI-04-2 With input information

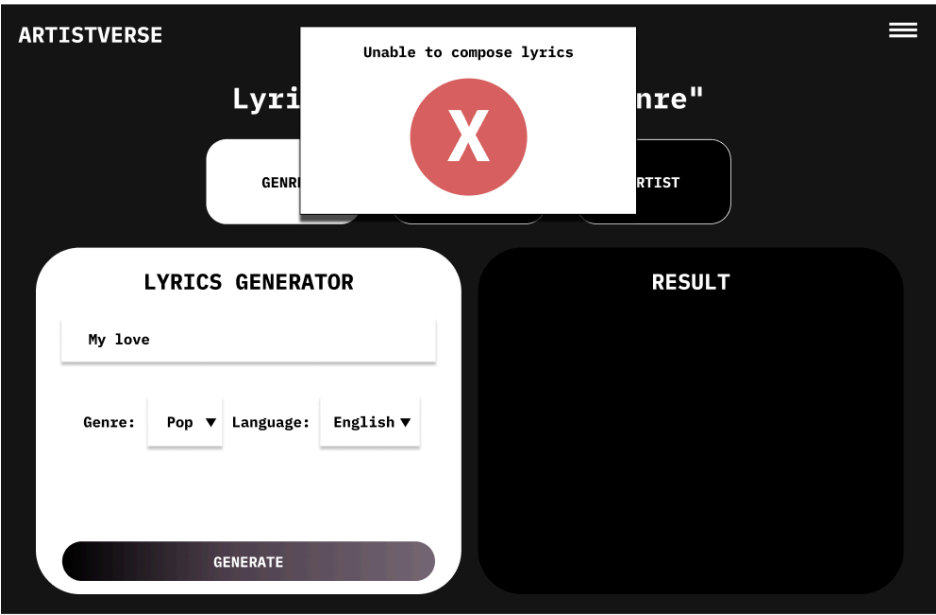


UI-04-3 With display the created lyrics

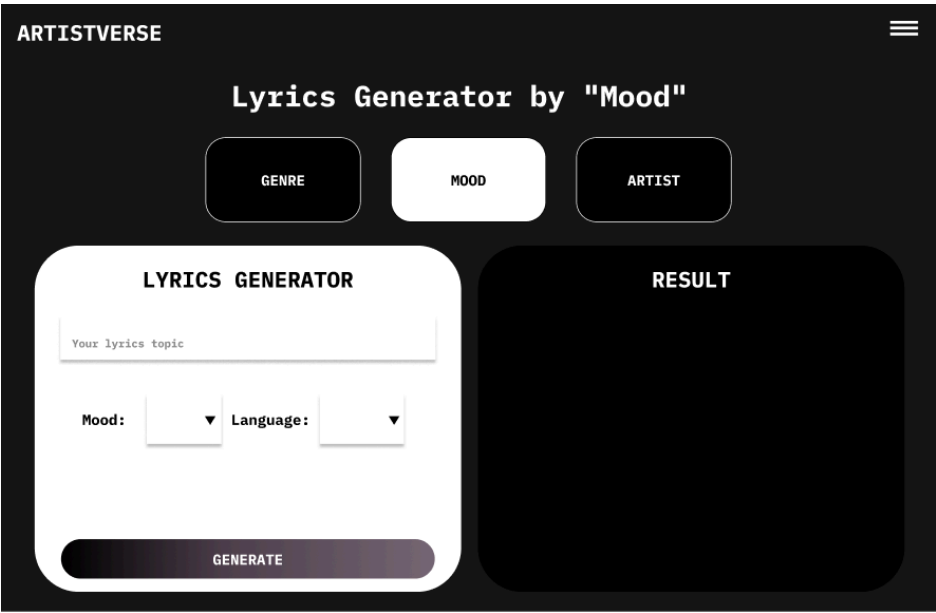


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UI-04-4 With causing by no internet connection

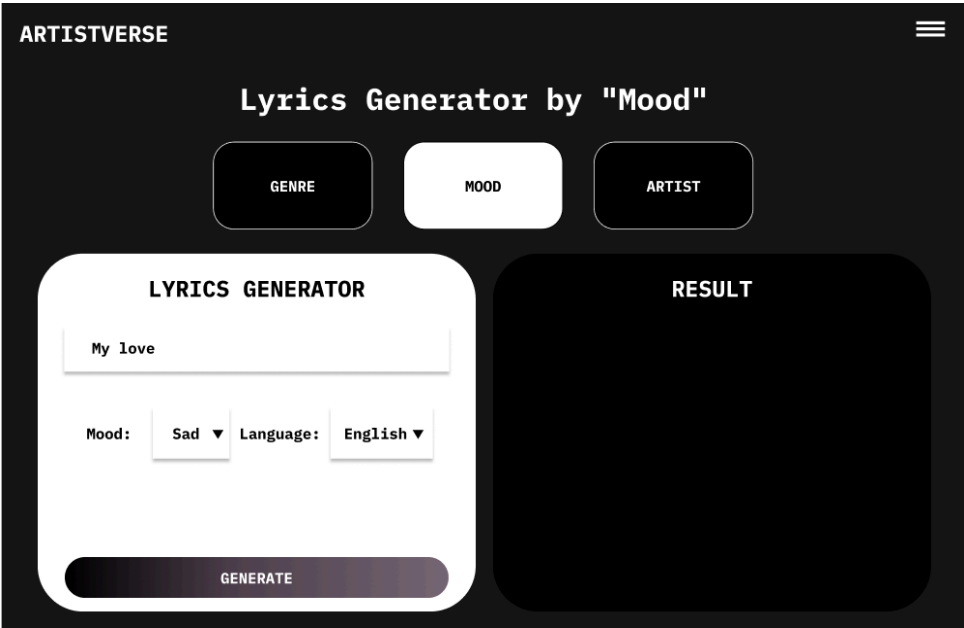


UI-05 Lyric Generator by Mood page

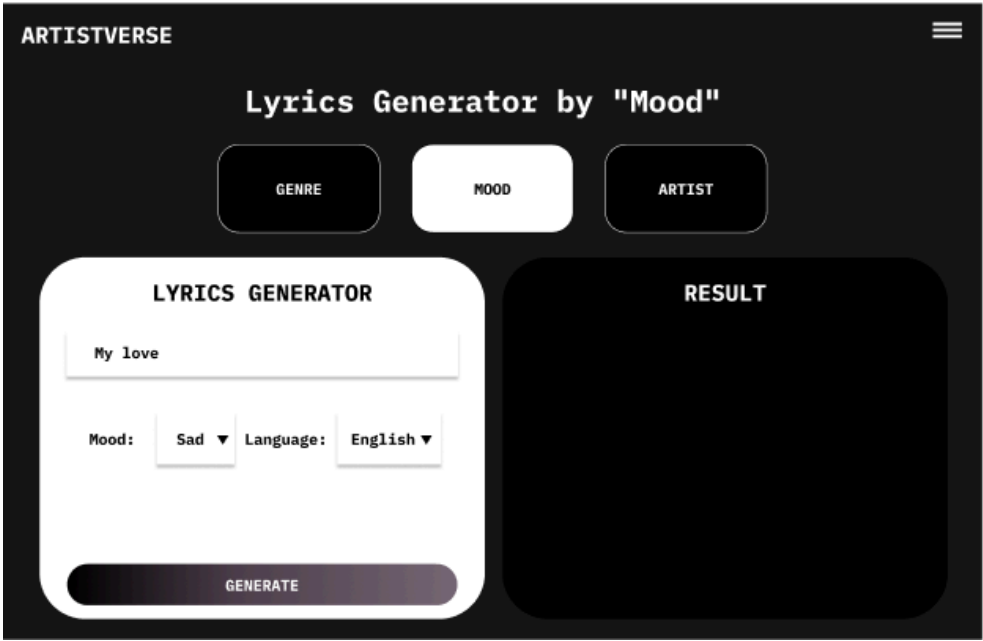


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UI-05-1 With input information

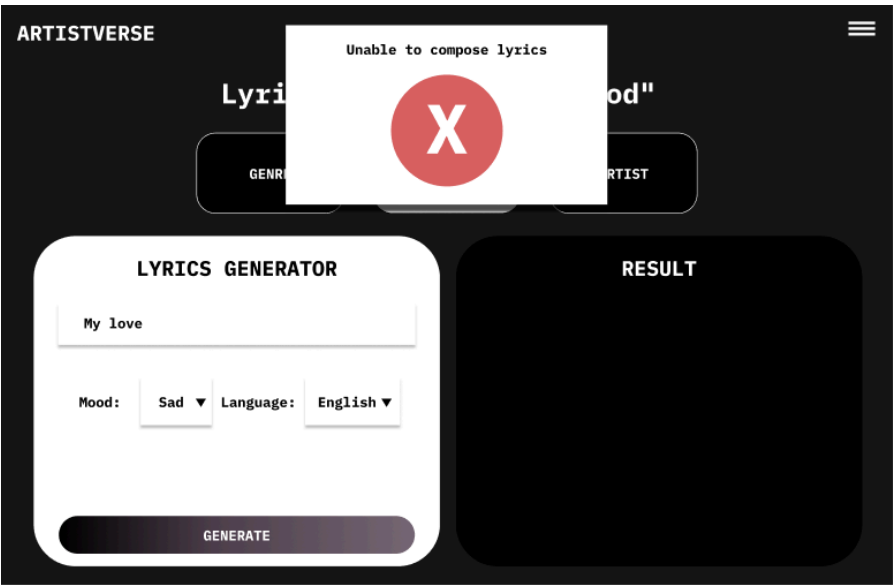


UI-05-2 With invalid input

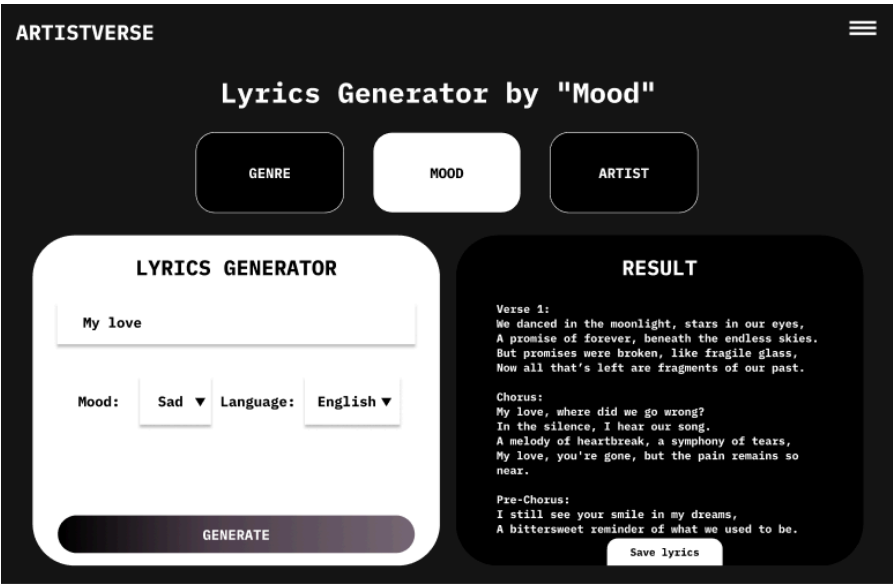


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UI-05-3 With causing by no internet connection



UI-05-4 With display the created lyrics



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UI-06 Lyric Generator by Style Artist Page

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

Your lyrics topic

Artist: Language:

GENERATE

RESULT

UI-06-1 With invalid input

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

Please input a lyric topic

Artist: Language:

Please select Artist

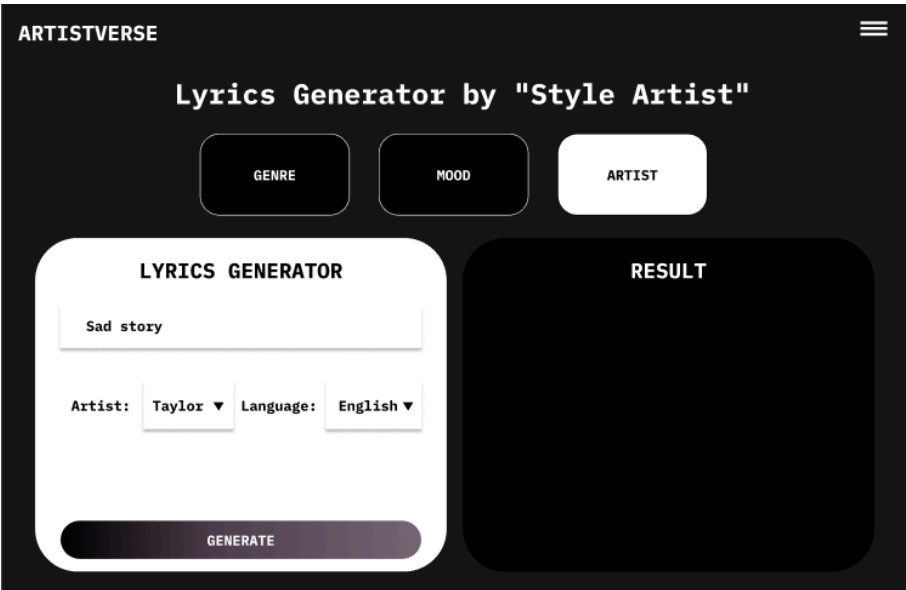
Please select language

GENERATE

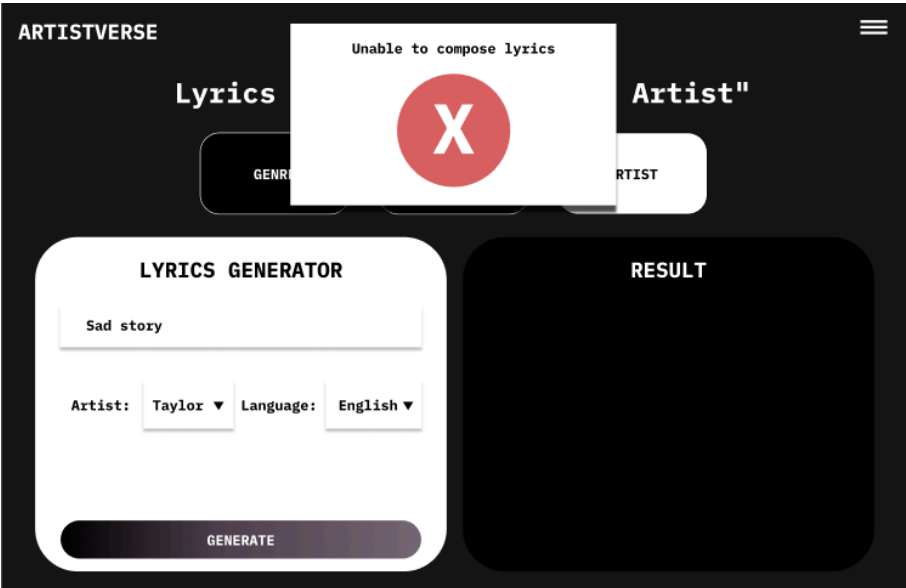
RESULT

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UI-06-2 With input information



UI-06-3 With causing by no internet connection



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UI-06-4 With display the created lyrics

ARTISTVERSE

Lyrics Generator by "Style Artist"

GENRE

MOOD

ARTIST

LYRICS GENERATOR

Sad story

Artist: Taylor ▼ Language: English ▼

GENERATE

RESULT

Verse 1:
We met in the summertime, laughter in the air
Your eyes were like a sunrise, lighting up
everywhere
We danced under the moonlight, whispers in the
dark
I thought we had a fairytale, but you left me
with a scar
Chorus:
You said forever, but you meant goodbye
Left me here in pieces, wondering why
The love we had was just a lie
Now all that's left are tears I cry

Save lyrics

UI-07 Lyric Generator page with save lyrics

ARTISTVERSE

Lyrics Generator by "Genre"

GENRE

MOOD

ARTIST

Lyrics Generator

My love

Genre: Pop ▼ Language: English ▼

GENERATE

RESULT

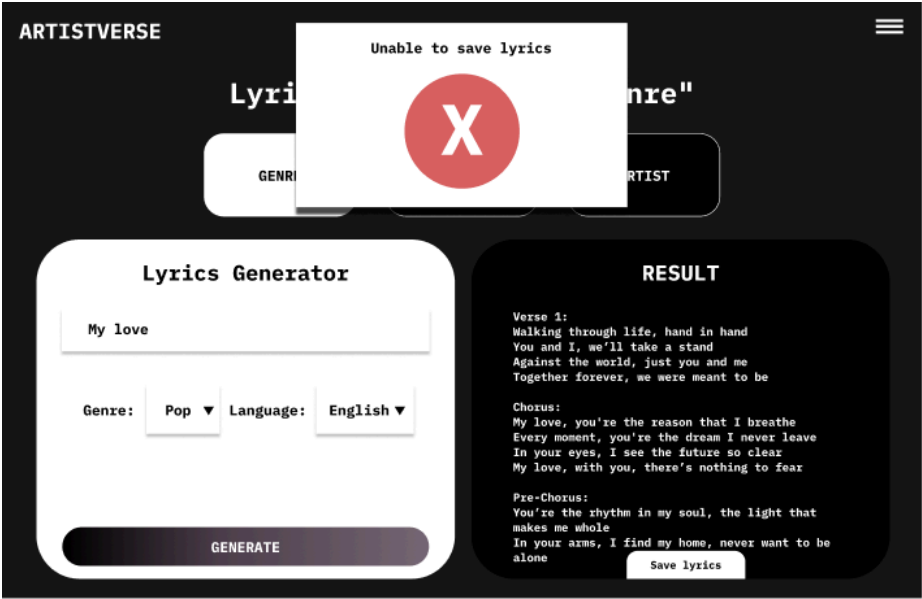
Verse 1:
Walking through life, hand in hand
You and I, we'll take a stand
Against the world, just you and me
Together forever, we were meant to be
Chorus:
My love, you're the reason that I breathe
Every moment, you're the dream I never leave
In your eyes, I see the future so clear
My love, with you, there's nothing to fear
Pre-Chorus:
You're the rhythm in my soul, the light that
makes me whole
In your arms, I find my home, never want to be
alone

Save lyrics

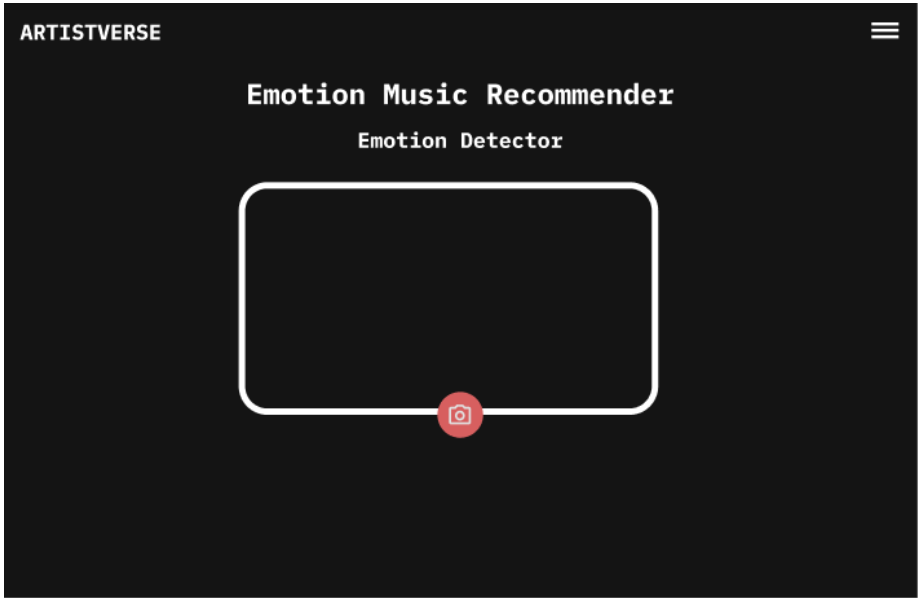
Lyrics.txt
"progress 13.23kb / 23.41kb"

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UI-07-1 With Unable to save lyrics causing by no internet connection

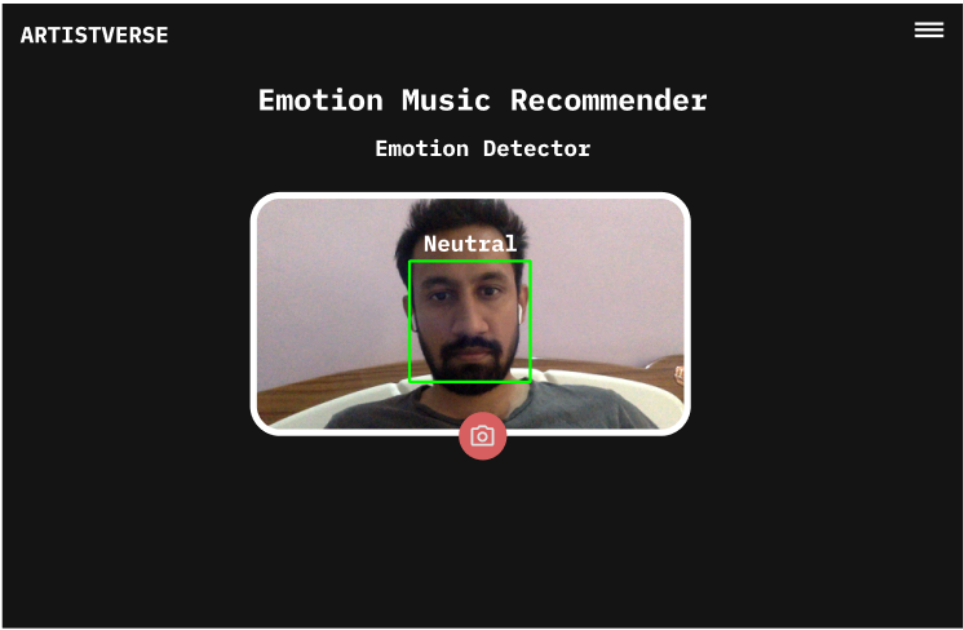


UI-08 Camera mood page

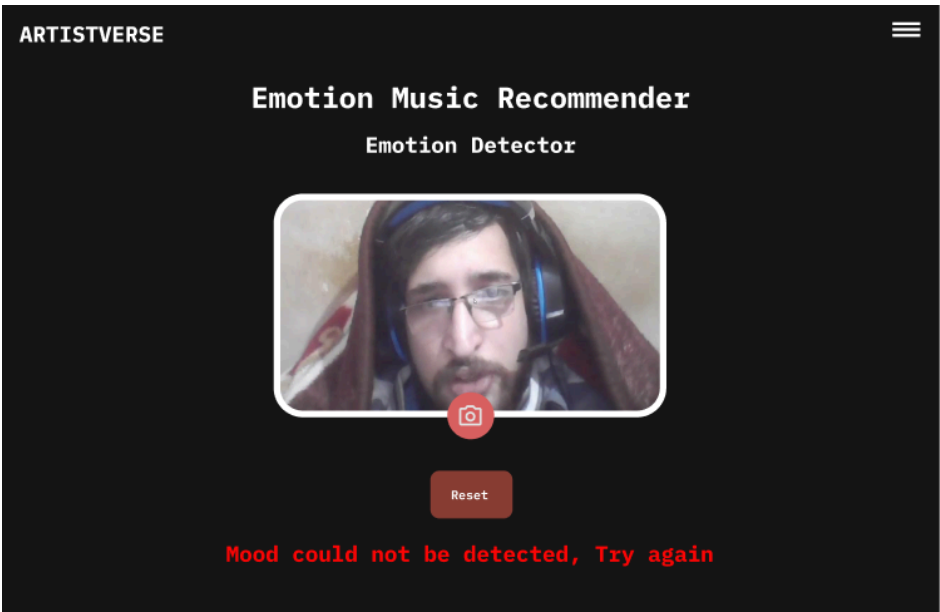


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UI-08-1 With camera view

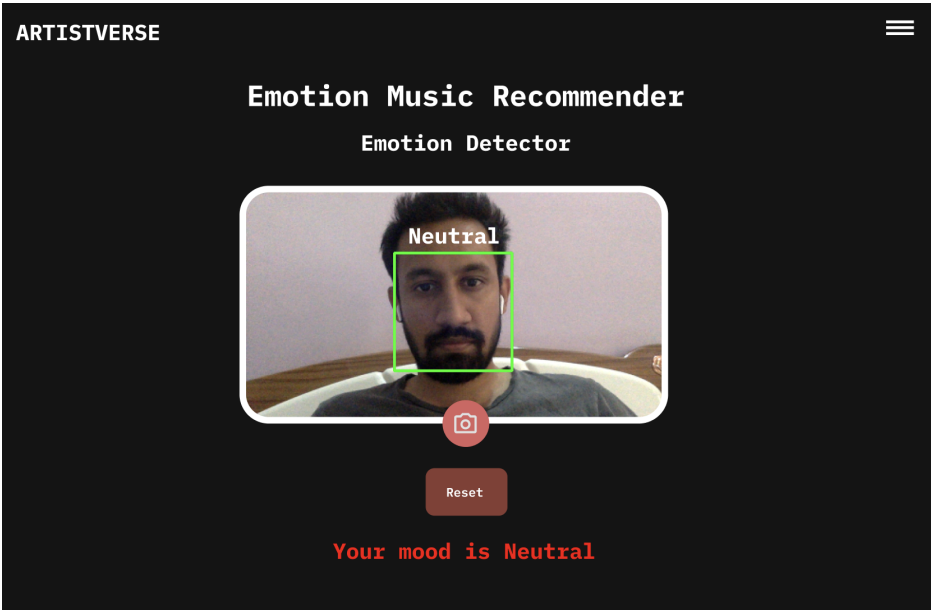


UI-08-2 With system unable to determine the mood



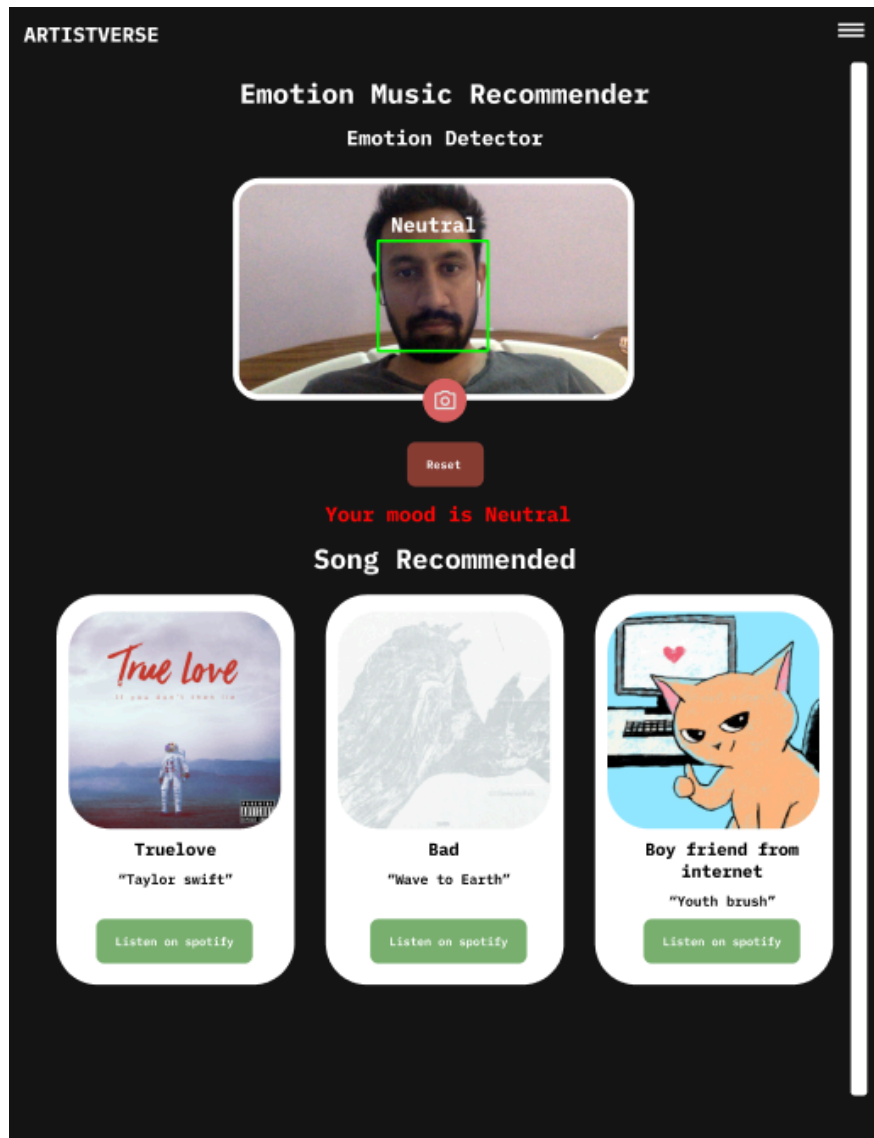
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UI-08-3 Camera mood page with detected mood



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UI-08-4 With song recommended and song choosing

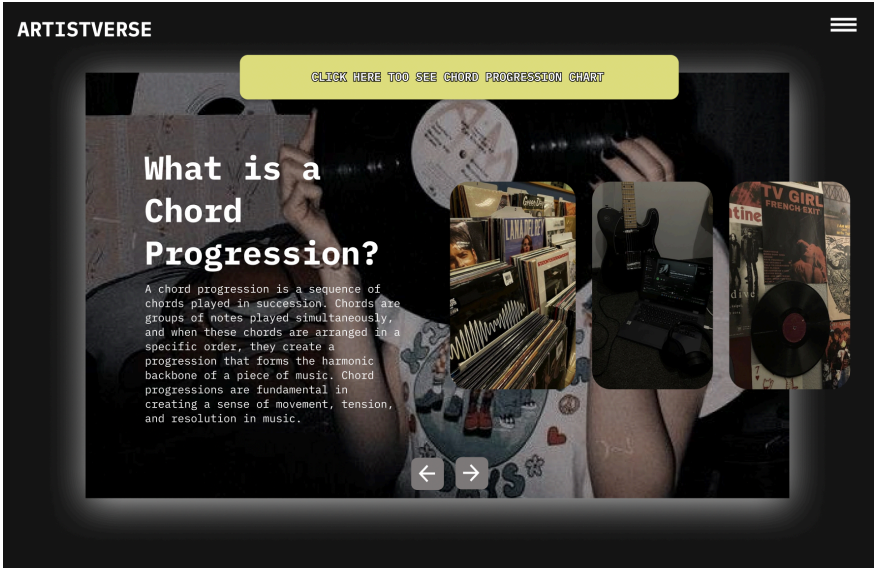


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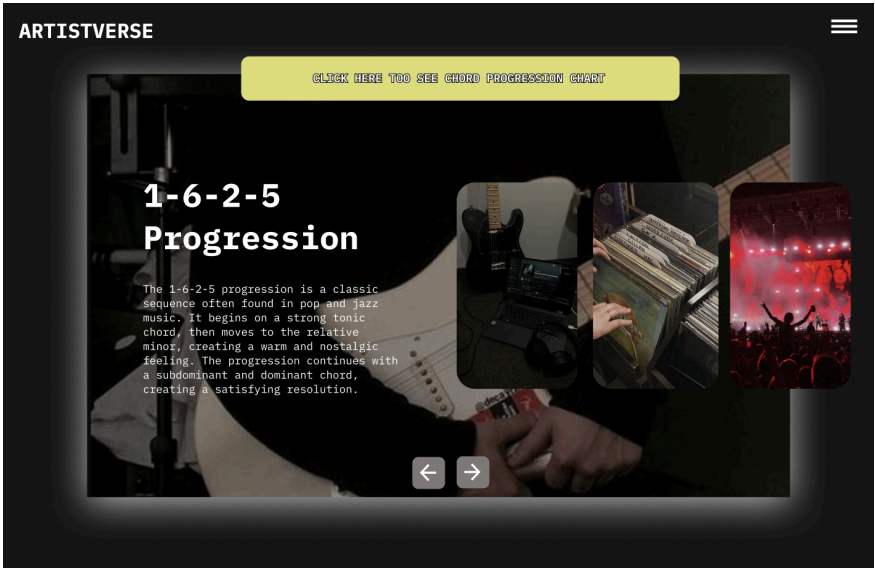
4.2 Progress II

4.2.1 Web Application UI

UI-09 Chord Progression page



UI-09-1 Chord Progression page when click next button



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UI-10 Chord Progression Table page

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

UI-10-1 With selected progression

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

Selected Chord Progression: 2-5-1-6 in Key C

Dm

G

C

Am

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UI-10-2 With highlights the chord in the progression with an orange colour as the sound it is played.

ARTISTVERSE

Major Keys Chord Table

Key	Progression								1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C	Dm	Em	F	G	Am	Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D	Em	Gbm	G	A	Bm	C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E	Gbm	Abm	A	B	Dbm	D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F	Gm	Am	Bb	C	Dm	Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G	Am	Bm	C	D	Em	F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A	Bm	Dbm	D	E	Gbm	G#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-6-4-5	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B	Dbm	Ebm	E	Gb	Abm	A#dim

Selected Chord Progression: 2-5-1-6 in Key C

G

C

D

G

UI-11 AI-Assistance page

ARTISTVERSE

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

Type a message...

Send

Save Analyzed Lyrics

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UI-11-1 With invalid input

ARTISTVERSE

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

The lyrics field cannot be empty. Enter your lyrics to proceed.

Type a message...

Send

Save Analyzed Lyrics

UI-11-2 With User input

ARTISTVERSE

Welcome to AI Assistance

This interface provides music suggestions based on lyrics you provide.

Your lyrics

Suggestion

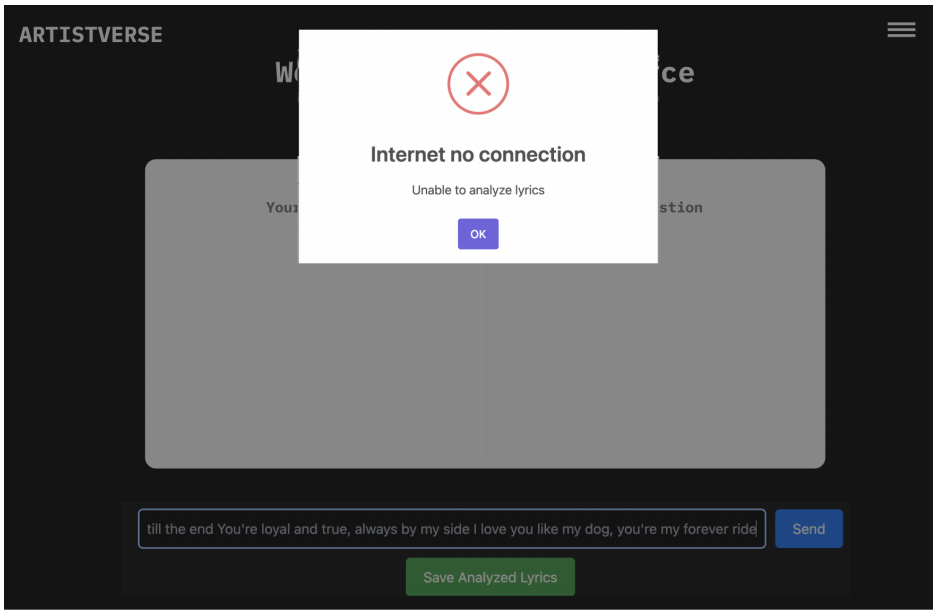
till the end You're loyal and true, always by my side I love you like my dog, you're my forever ride

Send

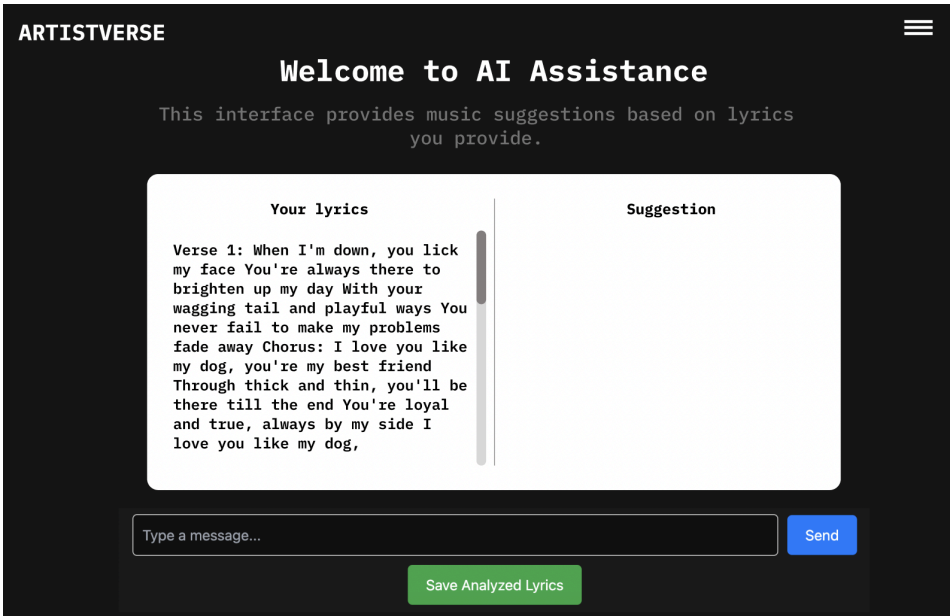
Save Analyzed Lyrics

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UI-11-3 Proceed without internet connection

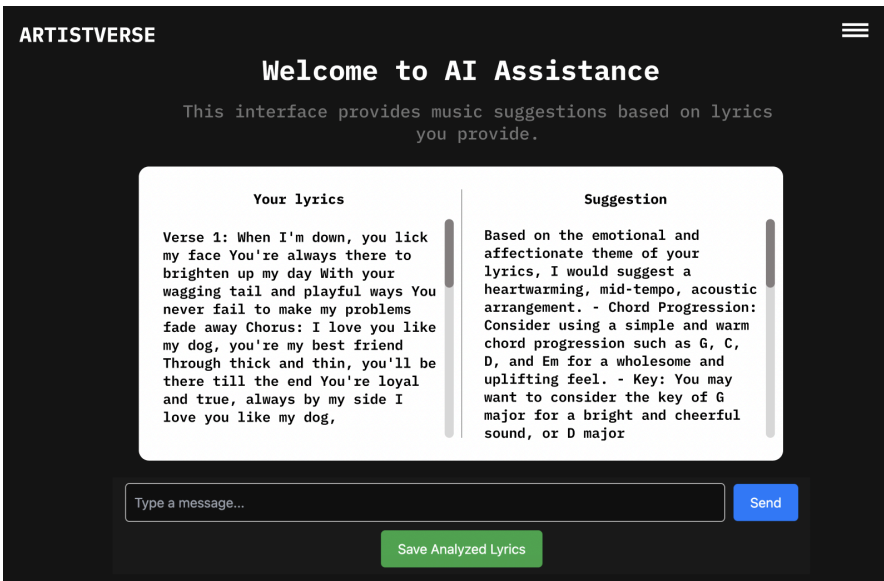


UI-11-4 With input after click button “Send”

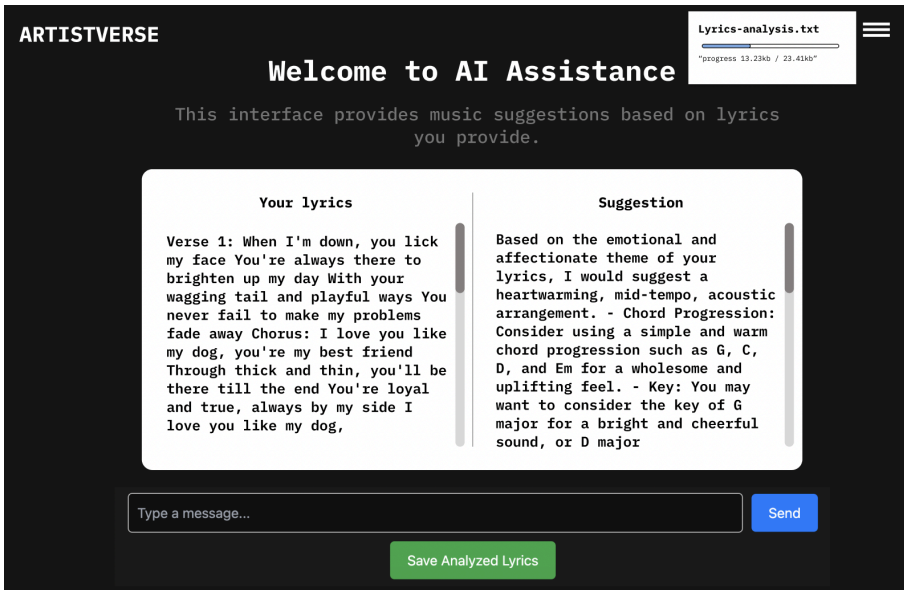


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UI-11-5 With suggestion analyzed

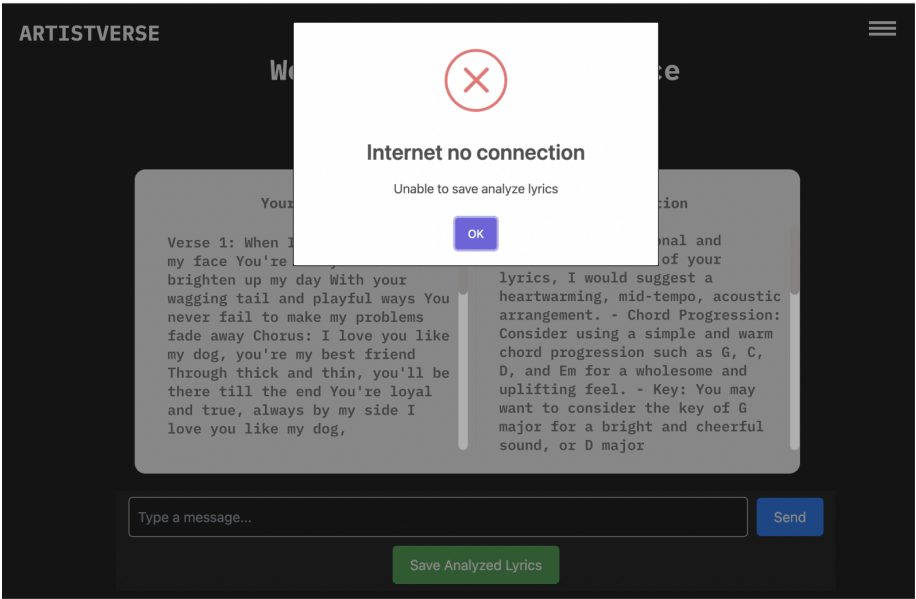


UI-11-6 With User click save analyzed lyrics



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UI-11-7 With no internet connection when clicking “Save Analyzed Lyrics”



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